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ORIGINAL RESEARCH

From unstructured text to structured reasoning: a hybrid knowledge graph for Indonesian sentencing analysis

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Abstract
Indonesian court decisions contain rich legal knowledge about how judges interpret statutes, assess evidence, and determine sentences, yet much of this remains hidden in semi-structured, inconsistent documents. Rule-based systems fail on lengthy narratives, while machine learning models overfit to noise, and large language models risk factual errors. This study introduces a hybrid framework that combines rule-based extraction for structured sections with a BERT-based pipeline for narrative text. This framework was developed using a dataset of 9,109 narcotics and corruption cases; from this corpus, seven legal experts manually annotated 200 decisions. A relevance filter cleaned the text before entity extraction using fine-tuned LegalBERT. The entity extraction pipeline achieved an average F1-score of 84.39% across both domains (86.1% for corruption cases and 82.3% for narcotics cases), and the resulting knowledge graph enabled sentence length prediction with 85% accuracy, i.e., a 56.9% point improvement over full-text baselines that reached only 26–28% despite 95% training accuracy. This notable performance gap suggests that structured graph representations may better capture legally meaningful patterns that unstructured text alone tends to miss. Our study introduces what we believe is the first legal knowledge graph of Indonesian court decisions.

Keywords Knowledge graph · Legal information extraction · Hybrid framework · Rule-based extraction · BERT · Indonesian court decisions · Sentencing prediction · Legal analytics

1 Introduction

Legal certainty in civil law systems depends not only on codified statutes but also on the consistency of judicial interpretation. From Weber's account of bureaucratic rationality to contemporary work in legal informatics, predictability emerges

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as a core attribute of legitimate adjudication (Ashley 2019; Sørensen and Torfing 2024). Indonesia, inheriting a Civil Law tradition from Dutch colonial administration, places primary authority in legislative codes, unlike Common Law systems that foreground precedent (Bhakti 2016; Butt and Murharjanti 2022; Lewis 2021). Yet codification cannot anticipate every factual configuration. Jurisprudence, therefore, operates as the bridge between abstract norms and concrete disputes (Simanjuntak 2019). Supreme Court cassation decisions in Indonesia form a crucial interpretive layer, connecting statutory principles to practice. They contain factual reconstructions, evidentiary references, and judicial reasoning that reveal how the law is applied (Butt 2019). Despite this richness, their analytical potential remains underutilized due to structural and linguistic limitations.

Unlike the standardized databases of the European Court of Justice or U.S. federal courts, Indonesian judicial decisions lack consistent metadata, formatting conventions, and semantic markup. Although substantively detailed, they are structurally fragmented, which obstructs systematic comparison (Bhakti 2016; Napitupulu, 2022; Van Opijnen et al. 2017). Supreme Court Circular Letter (SEMA)[1] No. 1 of 2022 introduced writing templates, but implementation has been uneven. The fact (fakta) section, describing prosecution narratives and evidentiary material, may range from a few hundred to over fifteen thousand words. Judicial style and case complexity shape its length. Extensive copy-pasting from indictments and testimonies produces redundancy and textual noise, sometimes dominating the document (Ginting, 2020). Our preliminary review confirms duplication, typographical inconsistencies, and even repeated verdict segments that obscure chains of reasoning.

Legal reasoning itself is often implicit. Judges cite statutory provisions without clearly articulating how proven facts satisfy legal elements (Butt 2021). This limits traceability. Linguistic heterogeneity further complicates analysis: Dutch-derived legal terminology, such as rechtskracht, and Arabic expressions, such as ta'zir, coexist with regional dialects within witness statements in a single judgment. Such diversity impedes tokenization and semantic parsing, posing challenges to both doctrinal scholars and computational systems. Despite these structural obstacles, Indonesia lacks a national framework for standardized digital court records.

Under these conditions, existing extraction systems perform inconsistently. Rule-based approaches handle structured segments effectively but struggle with narrative passages (Ariai et al. 2025; Chaitra et al. 2024; Kirsch et al. 2025). Deep learning models, while powerful, often overfit noisy or repetitive patterns instead of capturing legally meaningful relationships (Ariai et al. 2025; Chaitra et al. 2024). Large language models (LLMs) show promise for legal NLP but raise documented concerns regarding hallucination and reliability, particularly where precision is essential (Dahl et al. 2024; Deroy et al. 2025). While a direct empirical comparison with contemporary LLMs falls outside this study's scope, these architectural constraints motivate a framework that prioritizes auditability and schema-enforced extraction over end-to-end generative approaches. An effective system must therefore process semi-structured, inconsistent text while preserving interpretability and producing verifiable outputs.

[1] https://jdih.mahkamahagung.go.id/index.php/legal-product/sema-nomor-1-tahun-2022/detail.

In this study, structure refers to the explicit modelling of legally meaningful relationships rather than relying on implicit textual patterns. It captures institutional facts (issuing court and decision), core offense elements (act, actor, quantity, time, place), evidentiary links (evidence supporting specific claims), and statutory connections (legal provisions governing acts). This representation enables traceable reasoning, for example, linking a drug quantity under Article 114(2) of Law 35/2009 to its sentencing range, whereas unstructured models infer such relations only from textual co-occurrence. Structural heterogeneity refers to formatting variation across courts, while semantic heterogeneity denotes linguistic variation in the expression of equivalent legal concepts. The hybrid framework addresses both through differentiated extraction strategies.

This challenge is not merely technical. Persistent sentencing disparities, especially in corruption and narcotics cases identified by the Indonesian National Risk Assessment on Money Laundering, underscore the absence of structured judicial data (Pusat Pelaporan dan Analisis Transaksi Keuangan, 2022). Without systematic representations of statutory application or evidentiary weighting, it is impossible to evaluate whether similar cases receive similar outcomes, whether proportionality is respected, or which evidentiary forms most influence sentencing. These questions lie at the heart of fairness but remain inaccessible in an unstructured archival regime.

Normative legal theory emphasizes predictability, transparency, and coherence as pillars of adjudication (Alexy 2010; Prakken 2010). Knowledge graphs offer a means to operationalize these ideals. By modelling decisions as interconnected entities, defendants, acts, evidence, statutory articles, and defining relations such as performs, violates, or proven by, graphs render reasoning chains explicit. They enable quantitative consistency checks and qualitative exploration of interpretive patterns (dAmato et al. 2025; Kondo et al. 2025). By linking offense characteristics, evidentiary strength, and sentencing outcomes within a structured schema, they also facilitate proportionality assessment.

Recent advances in computational law illustrate this potential. Dugac and Altwicker (2025) employed LLMs to classify interpretive techniques in European Court of Human Rights judgments, demonstrating that generative models can detect textual and teleological reasoning while facing constraints in lengthy, noisy texts. Other research has applied argument mining and graph-based modelling to judicial corpora (Dong et al. 2024; Habernal et al. 2024). However, these studies typically rely on relatively standardized datasets. Indonesian decisions, by contrast, exhibit irregular formatting and linguistic diversity. Addressing this complexity requires methodological adaptation.

This study introduces a hybrid framework for constructing legal knowledge graphs from semi-structured Indonesian court decisions. By combining rule-based extraction with machine learning and applying relevance filtering to reduce textual noise, the approach improves extraction quality and enables scalable graph construction based on expert-annotated data. Structured representation significantly outperforms full-text baselines in sentence-prediction tasks, indicating that graph models better capture underlying legal logic than superficial textual patterns. The framework offers an interpretable and scalable solution for low-resource civil law contexts, transforming fragmented judicial archives into analyzable data. Although no direct com-

parison with contemporary LLMs is conducted, the design prioritizes auditability, efficiency, and data governance under institutional constraints rather than systematic benchmarking.

The remainder of this paper proceeds as follows. Section 2 reviews related work on legal document processing, situating our contribution within the shift from rule-based methods to hybrid frameworks and legal knowledge graphs. Section 3 outlines the methodology, including data scope, extraction pipeline, schema design, and experimental setup. Section 4 reports empirical results on extraction performance and sentencing prediction. Section 5 discusses implications for transparency, ethics, and cross-jurisdictional application. Section 6 concludes with limitations and directions for future research.

2 Background and related work

This section reviews prior work on legal document processing, examining the evolution from rule-based systems to machine-learning approaches, the emergence of hybrid frameworks, and the development of legal knowledge graphs. We position our work critically within this landscape and identify the gaps our research addresses.

2.1 Evolution of legal document processing methods

Computational analysis of legal documents has evolved through several methodological phases (Frankenreiter and Livermore 2020). Early work in the 1970s and 1980s focused on expert systems that encoded legal reasoning using handcrafted rules. Projects such as HYPO and CATO modelled argumentation using case-based reasoning. They demonstrated that structured representations could emulate aspects of legal reasoning, but at the cost of intensive manual knowledge engineering (Ashley 1990; Ashley et al. 2002; Bench-Capon 2017). These systems proved effective on limited, curated corpora but could not scale beyond them.

The rise of statistical NLP in the late 1990s and the early 2000s marked a shift toward corpus-driven methods. Schweighofer and Geist (2007) applied text mining to Austrian legal materials using frequency-based techniques to identify relevant cases, whereas Mochales and Moens (2007) developed template-based extraction for judicial decisions. Habernal et al. (2024) further extend binary classification into multi-class extraction of premises, conclusions, and issues from court opinions. These approaches improved automation but treated legal language as ordinary text. They ignored the argumentative structure that connects facts, rules, and conclusions. Because legal reasoning depends on relational logic rather than lexical proximity, bag-of-words methods fall short of capturing its essential structure.

Recent advances in deep learning have enabled the development of more context-aware approaches. Chalkidis et al. (2020) introduced LEGAL-BERT, a transformer model pre-trained on English legal corpora, including contracts, court cases, and legislation. It achieved significant gains over general-purpose BERT models, with an F1 score of 79.8% versus 68.2% for contract element extraction, confirming the value of domain-specific pretraining. Similarly, Zhong et al. (2018) used neural networks

to predict judicial outcomes in China, achieving 70–80% accuracy across multiple tasks, including charge classification and sentence prediction. Their work showed that when sufficient data are available, neural models can capture patterns of judicial reasoning. Yet purely neural approaches have two critical limitations. They require massive datasets; for example, Zhong et al. (2018) trained on 2.68 million cases, and they lack interpretability. As Deeks (2019) notes, black-box models are unsuitable in contexts where legal outcomes affect rights and liberties, since courts must understand the reasoning behind algorithmic decisions.

Most research has focused on Common Law jurisdictions, such as the United States, the United Kingdom, and India, or on data-rich Civil Law systems, such as the European Union and China. Indonesia remains largely unexplored despite presenting a distinct challenge. As a Civil Law country with limited digital infrastructure, its court decisions are plagued by inconsistent formatting, extreme redundancy where 20 to 60% of text is repetitive, and multilingual variation (Ginting, 2020). These conditions render the existing methods developed for standardized, high-resource settings inadequate, emphasizing the need for approaches that combine structural representation with robustness to noise, such as the hybrid knowledge graph framework proposed in this study.

2.2 Hybrid frameworks: combining precision and flexibility

Given the semi-structured nature of legal documents, hybrid approaches combining rule-based precision with the adaptive capacity of machine learning have become increasingly prominent (Akter et al. 2025; Siddiqi et al. 2025). This approach reflects the dual nature of legal reasoning: deductive inference from rules demands precision, while abductive reasoning from facts to explanations requires contextual flexibility (Lind 2022). Therefore, a computational system capable of mirroring this logic must integrate deterministic rule application with statistical pattern recognition.

Court decisions typically consist of both structured and unstructured sections. Structured components, such as case headers, verdicts, and closing statements, follow the prescribed templates. In Indonesia, SEMA No. 1 of 2022 specifies these formats, ensuring consistency in formal attributes, such as party names, statutory references, and the names of judicial officers. The unstructured fact section, however, contains a narrative core, i.e., descriptions of criminal acts, evidence, and testimonies that vary widely in length, tone, and detail. Understanding a decision requires interpreting both types of content within a legal framework guided by expert reasoning.

Hybrid and entity-aware approaches have recently gained attention in predictive legal AI (Benedetto et al. 2025; Dong et al. 2024). These studies use legal entities and graph structures to enhance judgment prediction and explanation, but remain focused on classification accuracy within homogeneous datasets. Our framework differs by emphasizing representational integrity, constructing verifiable legal knowledge graphs that preserve explicit logical and relational semantics rather than relying on latent embeddings. This approach aligns with Habernal et al., (2024), who call for integrating doctrinal reasoning models into computational pipelines, and extend such integration to under-standardized judicial texts. Dugac and Altwicker (2025) similarly applied LLMs to classify interpretative reasoning in judicial texts, iden-

tifying textual, systematic, and teleological patterns but noting the interpretability challenges of long and unstructured legal data. In contrast, our framework combines rule-based logic with neural learning to maintain transparency and scalability across heterogeneous corpora. We advance computational interpretation from reasoning classification to reasoning representation.

Hybrid legal NLP systems have also been successfully developed in other contexts. Waltl et al. (2017) used combined rule-based extraction and active machine learning to classify legal norms in German statutory texts, achieving F1-scores of up to 0.90 for certain categories such as objections, rights, and obligations. These metrics outperformed conventional supervised learning baselines. Similarly, Bach et al. (2019) applied a BiLSTM-CRF model to Vietnamese legal documents for reference extraction and achieved an F1-score of 95.35%. They demonstrated the effectiveness of combining traditional and deep learning methods in legal text processing. Their results are particularly relevant, given the linguistic and structural similarities between Vietnamese and Indonesian legal texts (Nguyen and Le 2020). Medvedeva et al. (2020) and Zhong et al. (2020) extended this approach to the ECHR decisions, combining rules for metadata extraction with machine learning for argument identification and demonstrating that hybrid systems can scale across thousands of cases while maintaining interpretability.

Our study is challenged by Indonesian court decisions that are rife with redundancy, with much of their content consisting of repeated indictments and copied witness testimony. This situation obscures the information most relevant to legal reasoning (Ginting, 2020). We also argue that legal discourse contains structural regularities such as citation syntax and evidentiary phrasing that transcend linguistic boundaries, proposing a practical pathway for developing legal NLP in lower-resource jurisdictions with limited domestic corpora (Ariai et al. 2025). A further limitation of the earlier hybrid systems is their narrow evaluation criteria. Most studies assess extraction accuracy but do not determine whether the results support substantive legal reasoning (Abdallah et al. 2023; Quevedo et al. 2024). Our research extends the hybrid paradigm in three key ways: by introducing multistage relevance filtering to manage extreme textual noise; by demonstrating the effectiveness of cross-lingual legal transfer for resource-limited jurisdictions; and by validating performance on substantive legal tasks rather than isolated extraction metrics.

2.3 Information extraction from legal documents

Information extraction from legal documents has evolved in several methodological stages. Effective segmentation, in which texts are divided into coherent parts, such as case summaries, arguments, verdicts, and citations, is a prerequisite for structured legal analysis and retrieval. Early approaches relied heavily on rule-based systems (Aumiller et al. 2021; Kalamkar et al. 2022). Solihin and Budi (2018), for example, used regular expressions to identify entities, such as case numbers, defendant names, and verdicts, in Indonesian court decisions. Their method achieved high precision (92%) in structured sections but failed to generalize to unstructured narratives, where linguistic variability and inconsistent formatting disrupted the application of the rules. The results highlighted a key limitation of rule-based extraction whereby rules

tailored to one judge's phrasing often fail when applied to another, reflecting the inherent stylistic diversity of legal writing.

To overcome this rigidity, machine learning models such as Conditional Random Fields (CRFs) have been introduced. Nuranti and Yulianti (2020) applied CRF to Indonesian legal documents and achieved a 78% F1-score for named entity recognition of statutory references and party names. However, CRF models still struggle to capture relational contexts, such as distinguishing when the same individual appears in different legal roles or when entities interact causally within complex narratives. Advances in deep learning have further improved entity and relation extractions. Ahmad et al. (2020) used neural models to classify legal entities and relations, thereby reducing reliance on manual feature design. Bhattacharya et al. (2019) demonstrated that neural networks for Legal Case Retrieval can automatically identify relevant case segments and semantic associations, outperforming traditional feature-based methods. These studies have confirmed that deep models can infer abstract legal patterns that deterministic rules cannot encode.

However, such methods may encounter significant challenges in low-resource and noisy settings. Machine learning models trained on Indonesian court decisions frequently overfit document-specific expressions, rather than learning transferable legal concepts. High redundancy encourages the models to rely on superficial correlations. For example, phrases that appear in both guilty and acquittal verdicts may correlate misleadingly with sentencing outcomes, distorting predictions. Our filtering approach addresses this limitation by removing redundant and procedural texts before training the model. This preprocessing ensures that models learn from substantive legal content rather than stylistic or structural noise, thereby improving generalization and preserving the integrity of the extracted legal knowledge.

2.4 Large language models for legal information extraction

Recent studies examine the use of large language models (LLMs) for extracting legal information. Deroy et al. (2025) show that GPT-4 and Claude, when guided by customized few-shot prompts, achieve over 90% accuracy on structured fields (e.g., case numbers, dates) but below 60% accuracy on complex tasks such as legal reasoning and precedent identification. More broadly, LLMs have demonstrated strong performance across legal NLP tasks, including contract analysis and outcome prediction (Dahl et al. 2024; Deroy et al. 2025; Marcos 2025). Instruction-tuned models (e.g., GPT-4, Claude, Gemini) can perform zero-shot or few-shot extraction when provided with schema definitions and examples.

However, high-stakes judicial extraction raises distinct concerns about deployment. First, generative LLMs are vulnerable to hallucinations, plausible but factually unsupported outputs, which are unacceptable when extracting legally consequential facts (Dahl et al. 2024). Even modest error rates can compromise downstream applications such as precedent retrieval or sentencing audits. Second, LLM reasoning remains opaque. Extracted entities or relations cannot be reliably traced to specific source sentences, undermining explainability requirements emphasized in legal scholarship (Deeks 2019; Marcos 2025). Courts and practitioners must be able to audit and verify AI outputs. Third, computational costs and privacy risks emerge

when processing large judicial corpora via external APIs. Scaling to hundreds of thousands of decisions is financially burdensome and may require transmitting sensitive case data to third-party servers (Fernandez et al. 2025; Samsi et al. 2023).

Our hybrid architecture addresses these constraints by combining rule-based extraction for structured sections (ensuring precision and full auditability) with fine-tuned, locally deployable BERT models for narrative text (providing quantifiable reliability via F1-scores and traceable sentence-level outputs). The system supports human verification, avoids transmitting external data, and reduces recurring API costs. While advances such as retrieval-augmented generation may mitigate some LLM limitations, systematic comparison of accuracy, cost, and auditability across architectures remains future work beyond the scope of this framework demonstration in low-resource civil law contexts.

2.5 Legal knowledge graphs

Legal knowledge graphs represent a convergence between symbolic artificial intelligence and modern NLP (Chen et al. 2024). Unlike general-purpose graphs, such as DBpedia and Freebase, they must capture domain-specific properties, including statutory hierarchies, temporal validity, jurisdictional scope, and evidentiary relationships (DeLong et al. 2023). Early efforts in this field emphasized rule representation (Santosh et al. 2024). LegalRuleML formalized the encoding of obligations, permissions, and prohibitions, whereas Hoekstra et al. (2009) proposed the Legal Case OWL Ontology (LCOO), modelling cases as structured events with participants and outcomes. However, these frameworks were developed for European case law, assuming a uniform structure and explicit ratio decidendi, features that are often absent in Indonesian judicial texts.

Recent research has demonstrated that knowledge graphs make legal information more transparent and interconnected. Pan et al. (2024) note their ability to render complex facts navigable, whereas Castano et al. (2022) show that they help practitioners trace conceptual links across judgments. Further research highlights how linking open legal data through graphs enables comparative studies across jurisdictions, breaking language and accessibility barriers (Filtz et al. 2021; Kaltenboeck et al. 2022). In domains in which interpretive precision must coexist with large volumes of text, such structured representations enhance retrieval, analysis, and decision support (Chen et al. 2024; Ershov 2023).

The applications of legal knowledge graphs are diverse. Leitner et al. (2019) used these to detect inconsistencies in Austrian administrative rulings, revealing similar fact patterns with divergent judgments. Kondo et al. (2025) demonstrated that graph-based representations of legal reasoning outperform purely textual methods in retrieving relevant legal provisions and capturing structured reasoning paths between facts, norms, and applications beyond surface-level text. Our study developed a scalable method for extracting legal knowledge graphs from noisy, semi-structured court documents. Focusing on Indonesian decisions, it transforms judicial texts into graph structures that connect defendants, prosecutors, judges, statutory references, case facts, evidence, and verdicts. The resulting graphs preserve the logic of judicial

reasoning, while enabling semantic search, trend analysis, and visual exploration of decision patterns.

3 Methodological framework and design rationale

Our methodological approach is guided by three principles shaped by the nature of Indonesian court decisions. The first concerns structural heterogeneity. Indonesian legal documents contain both highly standardized and highly variable sections (Ginting, 2020). Case metadata and verdict formulas follow SEMA templates with stable linguistic patterns, while the facts section varies by clerk style, case complexity, and evidentiary composition. Because no single extraction technique performs well across this spectrum, rule-based methods are used for structured components due to their precision, and neural models are applied to narrative sections where contextual understanding is essential (Abrar et al. 2024; Tritto and Ponce 2025).

The second principle is the central role of domain expertise. Legal texts contain specialized reasoning that general NLP models often fail to capture (Marcos 2025). Important distinctions such as the evidentiary weight of statutory cross-references, the hierarchy of witness testimony, and the separation between actus reus and mens rea require doctrinal understanding. Legal experts, therefore, contribute not only annotations but also design decisions regarding which entities and relationships should be modeled.

The third principle concerns representation quality. Extraction is meaningful only when it supports legal reasoning. Our knowledge graph is validated through an extrinsic task, sentence length prediction, which operationalizes proportionality as a core principle of criminal law (Li et al. 2024). This ensures that the representation captures substantive legal relationships rather than superficial textual patterns. Guided by these principles, the hybrid methodology defines the data scope, identifies relevant features, performs rule-based and neural extraction, and establishes legally coherent relationships among entities.

3.1 Data collection and scope

Court decision documents were obtained from the official website of the Supreme Court of Indonesia (Mahkamah Agung Republik Indonesia), which provides publicly accessible records of judicial rule. Each entry includes metadata such as the document URL, case number, province, court of origin, case level, classification, verdict (length of punishment), document status, and downloadable PDF file. Two datasets are used in this study.

The first is the Indo-Law dataset, which comprises preprocessed XML files containing 22,630 first-instance criminal decisions from West Java, Central Java, East Java, Jakarta, and Yogyakarta. All decisions obtained a permanent legal force (inkracht).[2] Each XML file is structured into key components, i.e., kepala putusan (document header), identitas terdakwa (defendant identity), riwayat penahanan

[2] https://github.com/ir-nlp-csui/indo-law accessed on 23 February 2025.

(detention history), fakta (case facts), pertimbangan hukum (legal reasoning), amar putusan (verdict), and penutup (closing). The dataset adheres to three inclusion criteria: the decision must be a first-level criminal ruling issued by a district court, must have attained permanent legal force, and must have a corresponding PDF document available online.

Indonesian court decisions are written by legal professionals and follow strict doctrinal standards. The structure of Indonesian court decisions has historically been based on Dutch colonial procedural law (Herziene Inlandsch Reglement-HIR) and (Rechtreglement voor de Buitengewesten-RBG), which comprises: header, identity, case in dispute, legal reasoning, and decision. Following the enactment of Act No. 14 of 1970 concerning the Criminal Procedure Code (KUHAP) in the post-colonial era, the colonial designation of court decisions, which referenced the king, was altered to invoke the name of God. No significant alterations to the court decision structure have occurred to far, and it continues to adhere to the stringent framework outlined in the KUHAP. From a computational perspective, however, Indonesian court decisions contain substantial textual redundancy. Much of this redundancy arises from institutional requirements to reproduce indictments and witness testimonies verbatim in the fakta (case facts) section. In a typical narcotics judgment containing about 1,800 words, more than half of the section consists of boilerplate material. This includes roughly 200 words of procedural preamble that repeat indictment headers, about 500 words of formulaic witness testimony marked by recurring phrases such as "berdasarkan keterangan saksi yang dibacakan di persidangan," and approximately 300 words enumerating physical evidence items. The core of the legal reasoning is found in the remaining 800 words, which describe the criminal act, the evidence presented, and the defendant's role.

Although these repetitive segments are necessary for evidentiary completeness and legal formality, they contribute very little to the analytical determination of culpability and sentencing. Our relevance-filtering method, therefore, focuses on identifying and removing this institutional boilerplate while preserving the narrative portion that contains the legally meaningful information required for accurate entity extraction. The presence of redundancy in Indonesian decisions should not be interpreted as a lack of legal quality. The judgments maintain doctrinal rigor comparable to those in European or American courts. The computational challenge lies in separating substantive legal reasoning from mandated repetition, a challenge that does not typically appear in standardized repositories such as the ECHR, where decisions are already curated for analytical work.

The second dataset comprised the PDF documents themselves, with 702 files successfully downloaded. These were used to develop and validate the PDF-to-XML extraction pipeline detailed in Appendix A. The files correspond directly to entries in the Indo-Law dataset and were retrieved using the provided URLs. However, not all original PDFs remain available on the Supreme Court website, as some have been removed or relocated due to updates to the court's document management system. As shown in Fig. 1, the National Risk Assessment (2022) identifies corruption, narcotics, and tax evasion as predicate offenses most strongly associated with money laundering, providing empirical justification for our offense-type focus.

NO	TYPES OF PREDICATE CRIME	ML THREAT LEVEL	ML VULNERABILITY LEVEL	ML CONSEQUENCE LEVEL	ML LIKELIHOOD LEVEL	RISK LEVEL	ML RISK CATEGORY
1	Corruption	9,00	9,00	9,00	9,00	9,00	High
2	Narcotics	7,80	5,88	8,02	7,65	7,24	High
3	In Taxation Sector	6,92	3,90	8,28	6,73	6,74	Medium
4	In Banking Sector	6,25	6,00	7,16	6,90	6,18	Medium
5	In Forestry Sector	4,29	7,50	6,11	6,28	5,19	Medium
6	Fraud	6,37	5,59	5,51	6,86	5,14	Medium
7	In Environment Sector	4,46	7,03	6,01	6,26	5,12	Medium
8	Bribery	5,51	6,58	5,11	6,68	4,81	Low
9	Embezzlement	5,41	3,73	5,29	5,93	4,57	Low
10	Gambling	4,86	3,10	5,20	5,51	4,32	Low
11	Psychotropic	4,61	5,90	4,71	6,06	4,31	Low

Fig. 1 National risk assessment on money laundering 2022

To ensure analytical consistency, the dataset is refined using two constraints. The first is offense-type restriction. Guided by the National Risk Assessment on Money Laundering (2022), which identifies corruption, narcotics, and tax evasion as the highest risk predicate offenses, this study focuses on narcotics and corruption cases. Unlike more heterogeneous offenses such as assault or fraud, these two categories provide structured statutory definitions suitable for comparative analysis. Narcotics cases, governed by Law 35/2009, involve measurable quantities and concrete statutory elements, such as possession or trafficking. Corruption cases governed by Law 31/1999 entail relational elements such as position, authority, benefit, and intent. Together, they tested the model's ability to handle both tangible and abstract legal reasoning.

The second constraint concerns the completeness filtering. To maintain data quality, records were selected based on the completeness of the critical sections, particularly the fakta (facts) and amar putusan (verdict) components. Of the initial 22,630 records, 9,109 were retained after excluding incomplete cases, procedural dismissals, and appellate affirmations. This filtering ensured that the final dataset comprised first-instance criminal decisions that included full factual narratives and substantive judicial reasoning, the doctrinally appropriate unit for studying judicial decision-making patterns. These constraints serve methodological and theoretical purposes. Focusing on narcotics and corruption enables the design of a schema that captures distinct dimensions of legal complexity: one anchored in measurable evidence and the other in interpretive reasoning. This balance provides a robust foundation for evaluating

whether hybrid extraction methods can be generalized across different types of legal structures, while maintaining fidelity to the principles of Indonesian criminal law.

The 9,109 retained decisions represent only a small fraction of Indonesia's first-instance criminal adjudication between 2015 and 2022. Based on the Supreme Court's annual reports, an estimated 487,000 narcotics cases and 12,800 corruption cases were decided nationwide during this period, suggesting the dataset captures roughly 1.3% of narcotics decisions and 22.5% of corruption decisions. Several structural factors explain this limited coverage and introduce predictable selection biases. Because the digital repository records only decisions that have reached inkracht (permanent legal force), cases that remain under appeal, have been reversed, or have been dismissed procedurally are excluded. This requirement tends to favor more serious matters that proceed fully through the appellate process. Digitization practices also vary widely across courts. Urban jurisdictions such as Jakarta, Surabaya, and Bandung publish decisions more consistently than rural or remote courts, which means the dataset likely reflects metropolitan judicial patterns more strongly than those of regions with limited technological capacity. The need to filter out 13,521 incomplete records, which represented nearly 60% of the initially retrieved documents, further narrows coverage toward courts with more standardized documentation. In addition, the geographic focus on five provinces, which together account for about 58% of the national caseload, leaves out major regions such as Papua, Sulawesi, and Sumatra that may follow different judicial trajectories.

Even with these limitations, several features support the validity of the study's methodological claims. When sentencing patterns are examined province by province, the distributions remain remarkably consistent relative to statutory ranges, suggesting that the dataset captures stable judicial reasoning rather than idiosyncrasies of a particular location. The extracted determinants of sentencing, including quantity thresholds in narcotics cases and the role of positional authority in corruption cases, also align with analyses in Indonesian legal scholarship, providing an external check on interpretive accuracy. Most importantly, the study's primary contribution is methodological. It shows that hybrid extraction techniques can construct legal knowledge graphs from noisy, heterogeneous judicial texts, demonstrating that this does not require full national representativeness. Accordingly, the empirical findings are framed as patterns within digitized, permanent-force decisions from major urban courts. Expanding coverage to additional provinces, integrating appellate decisions, and adopting stratified sampling across courts with differing administrative capacities would strengthen future claims to broader generalizability.

3.2 The hybrid extraction pipeline

Our pipeline operationalizes the dual nature of legal documents by strategically allocating methods. A hybrid methodology was used to extract information and relations from court decisions. The process can be divided into two modules. Figure 2 illustrates the complete hybrid extraction pipeline, showing how rule-based and machine learning modules integrate to process structured and unstructured sections.

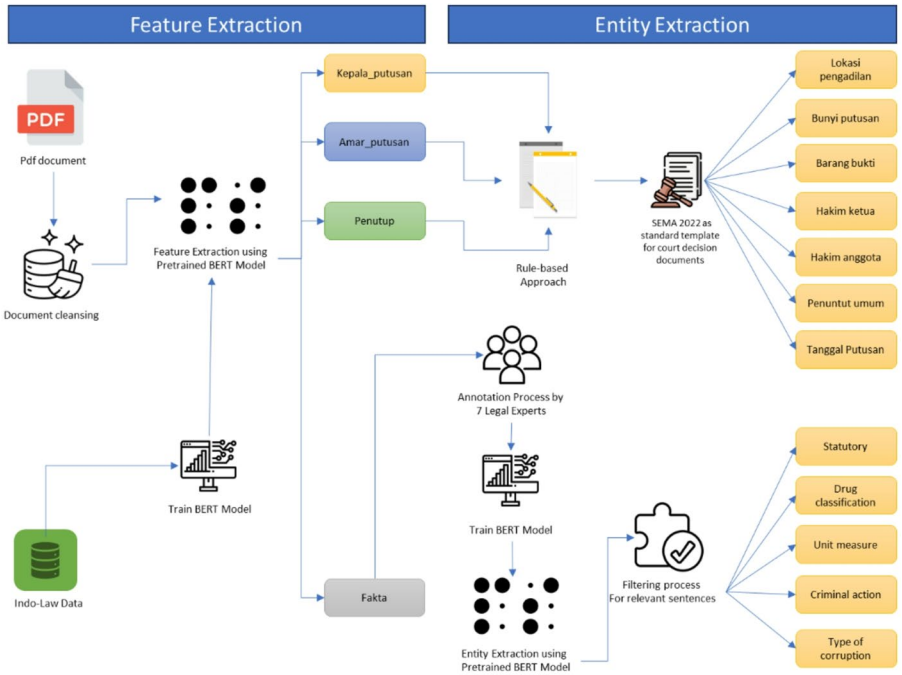


Fig. 2 Entity extraction process

3.2.1 Module 1: Rule-based extraction for structured sections

The kepala_putusan (document header), amar_putusan (verdict), and penutup (closing) sections follow SEMA 2022 templates with predictable marker phrases. Court decision documents can be classified as structured or unstructured. The structured sections follow templates defined by institutional requirements. SEMA No. 1 of 2022 specifies standard formats for specialized legal terminology, including statutory regulations, the names of parties involved, and those of judicial officers.

Extraction targets:

- From kepala_putusan: court location (lokasi_pengadilan), case number (nomor_dokumen). Rationale: Institutional identifiers enable provenance tracking and jurisdictional analysis. The court location allows the examination of regional sentencing patterns, whereas case numbers ensure unique identification and enable linking decisions across databases.
- From amar_putusan: verdict statement (bunyi_putusan), defendant name (terdakwa), charged offense (tindak_pidana), and evidence enumeration (barang_bukti). Rationale: The verdict section contains the court's final determination, the legally binding outcome. Extracting this information precisely is critical because any error in verdict extraction can misrepresent judicial decisions in downstream analytics.
- From penutup: decision date (tanggal_putusan), chief judge (hakim_ketua), mem-

ber judge (hakim_anggota), prosecutor (penuntut_umum), court clerk (panitera). Rationale: These entities enable analysis of judicial panels and prosecutorial patterns. Do certain judges demonstrate consistent sentence patterns? Do prosecutors achieve different conviction rates? Temporal information allows tracking of how judicial interpretations evolve over time.

The rules use regular expressions with marker phrases. These rules were defined based on SEMA 2022 as the standard template for court decision documents. For example, verdict statements begin with marker phrases such as "Mengadili" or "Menyatakan" and end with entity mentions like "terdakwa" or "barang bukti." However, in real situations, some documents do not comply exactly with the standard format. Therefore, we set multiple alternative marker words at the beginning and end of each section. This method achieved > 90% precision with minimal false positives.

These sections encode the institutional requirements. They represent bureaucratic facts that appear in every decision regarding procedural validity. Their templatic nature makes the rules appropriate. Rules provide high-precision, human-verifiable outputs, and require no training data. A 10% failure rate occurs when documents deviate from SEMA templates (using "Memutuskan" instead of "Mengadili"). These failures are acceptable given the high precision of compliant documents, and manual review can efficiently capture errors because rule-based extraction produces transparent outputs that make failures immediately identifiable.

To provide full procedural transparency and clarify the specific data transformations within the 'black box' of the pipeline, Table 1a defines the input units, content types, and resulting outputs for each major stage of the hybrid framework (see Table 1).

3.2.2 Module 2: Machine learning-based extraction for unstructured facts

The fakta (case facts) section of Indonesian court decisions contains the prosecutor's narrative of the criminal act, evidence, and witness's testimony. It is unstructured, stylistically inconsistent, and often exceeds 15,000 words in number. Understanding this requires legal interpretation rather than linguistic pattern matching. To process this complexity, we developed a three-stage machine-learning pipeline: expert annotation, relevance filtering, and entity extraction.

In the first stage, seven legal professionals specializing in narcotics and corruption annotated 200 decisions (100 per offense type) using Doccano and Ngrok. Their

Table 1 Pipeline stage mapping and data transformation

Pipeline Stage	Input Unit	Input Content	Output
Module 1 (Rule-based)	XML/PDF Chunk	Case Headers, Verdicts, and Closings	Structured Triples (Institutional Facts)
Module 2–2 (Relevance Filter)	Individual Sentences	Raw Factual Narrative (Unstructured)	Binary Label (Relevant/Irrelevant)
Module 2–3 (Entity Extraction)	Filtered Sentences	Legally Salient Facts (Substantive Content)	Legal Entities and Relations

annotations capture legally consequential entities (e.g., criminal_act, evidence, ACT (statutory article), drug_classification, unit_measure, and position_profile) each defined according to doctrinal categories such as actus reus, evidentiary standards, and aggravating or mitigating factors. Objective entities such as case numbers and statutory articles achieved high inter-annotator agreement (0.98 and 0.94), whereas interpretive ones, such as criminal_act and type_of_corruption, scored lower (0.52 and 0.47). This variation reflects genuine legal ambiguity: objective facts admit consensus, whereas interpretive judgments vary with context. This pattern captures the epistemic nature of law, where some facts are determinate, and others are inherently open to multiple readings.

The second stage addresses textual noise. Indonesian court decisions contain extensive procedural boilerplates and repetition, which confuses extraction models. Direct entity extraction on unfiltered text yielded only a 67% F1-score. Using a fine-tuned IndoBERT classifier trained on annotated sentences, we distinguished legally relevant text from irrelevant text. Because the relevant content accounted for fewer than 2% of all sentences, we applied the Synthetic Minority Oversampling Technique (SMOTE) to balance the dataset. The resulting model achieved 99.65% accuracy, perfect recall, and the removal of redundant material without losing meaningful content. Filtering raised downstream entity extraction performance from 67 to 84.39%, a 17.39-point gain that validates the principle of legal salience: only a small fraction of legal text carries substantive value. This approach directly addresses the redundancy endemic to Indonesian court writing, in which procedural phrasing often outweighs factual substances.

In the final stage, fine-tuned LegalBERT, an English-domain legal BERT model (Chalkidis et al. 2020), distinct from the IndoBERT model used in filtering and segmentation, is applied for entity extraction. LegalBERT is used here rather than IndoBERT because entity extraction benefits from cross-lingual legal-domain pretraining, as discussed below, whereas filtering and segmentation benefit from Indonesian-language-specific pretraining. Although trained on English corpora, LegalBERT effectively captured cross-lingual patterns in legal discourse (i.e., standardized citation forms, evidentiary structures, and hierarchical reasoning) that appear across Civil Law systems. Fine-tuning has adapted these structural regularities to Indonesian surface forms. The model used an AdamW optimizer with a learning rate of 2e-5 and a five-token context window on either side of each entity, optimal for short, clause-level phrasing typical of legal text. The corruption model achieved an average F1-score of 86%, whereas the narcotics model achieved 83%. Lower performance for position profile in narcotic cases (61.3%) reflects genuine legal irrelevance of occupational roles in drug offenses rather than modeling weakness.

LegalBERT outperformed IndoBERT by 6–8% across domains. This confirmed that domain-specific pretraining on legal corpora transfers more effectively than language-specific general training. This finding has broader implications for lower-resource jurisdictions, i.e., cross-lingual transfer from high-resource legal models is more efficient than building new models from general text. Overall, the pipeline integrates legal expertise, linguistic precision, and machine learning to transform verbose factual narratives into structured, interpretable data. Expert annotation grounds the model in legal reasoning, filtering isolates substantive content of isolates, and cross-

lingual fine-tuning enables robust extraction with limited local data. The result is a replicable method to convert unstructured judicial narratives into machine-readable representations that support sentencing prediction, consistency analysis, and interpretable legal AI development.

3.2.3 Annotation protocol and guidelines

The annotation was conducted by seven legal practitioners with over five years of criminal law experience from a national financial intelligence institution in Indonesia. None had prior NLP or machine learning training, ensuring that the labels reflected legal judgment rather than technical bias. All annotators used a guideline document outlining each entity type with doctrinal definitions, statutory references, and illustrative examples. For instance, criminal_act was defined as 'the physical element of the offense (actus reus) as described in the prosecutor's narrative or court's factual findings, excluding legal characterization or conclusions.' Examples distinguished factual descriptions ('memberikan uang sejumlah Rp 50 juta kepada pejabat X'—'giving 50 million rupiah to official X') from legal characterizations ('melakukan suap aktif'—'committing active bribery'). Similarly, evidence entities required material items (physical objects, documents, testimonies) explicitly referenced by the court as supporting factual findings, excluding procedural mentions (e.g., 'evidence labeled Exhibit A-1' without description of its content or evidentiary role). For relevance filtering, sentences were labeled binary: 'relevant' if they contained (a) criminal act descriptions with specific conduct details, (b) evidentiary items with their probative connection to facts, (c) statutory references applied to established facts, or (d) aggravating/mitigating factors affecting sentencing. 'Irrelevant' sentences included: procedural formulas ('Menimbang bahwa terdakwa telah diperiksa dalam keadaan sehat…'—'Considering that the defendant was examined in healthy condition…'), detention chronology without substantive reasoning, boilerplate citations of procedure codes without application to case facts, or verbatim repetition of indictment text without judicial commentary.

A double-blind protocol was used: each of the 200 documents was independently annotated by two reviewers, with discrepancies flagged for expert adjudication. Pre-adjudication agreement was high, with κ scores broadly within the 'substantial' to 'near-excellent' range commonly used in annotation studies. The 200-document annotated corpus, evenly split between narcotics and corruption cases, was partitioned 70/15/15 for training, validation, and testing of the entity extraction models, with category balance maintained across splits. Note that this split applies exclusively to the annotated subset used for supervised entity extraction; the split for the full 9,109-document corpus used in sentence length prediction is described separately in Sect. 3.4. These aggregate κ values reflect overall pipeline-level agreement across all annotators and entity types; entity-type-specific and domain-level breakdowns, which reveal the epistemological distinction between objective and interpretive legal facts, are reported in detail in Sect. 4.3 and Table 11.

3.3 Knowledge graph schema and relationship extraction

The knowledge graph schema design is guided by three foundational principles of Indonesian criminal law doctrine. First, the principle of legality (nullum crimen, nulla poena sine praevia lege poenali—no crime, no punishment without prior penal law) requires that criminal liability must be anchored to specific statutory provisions. This motivates our subject_to_article relation, which explicitly links criminal acts to the legal articles that define and prohibit them. Second, the evidentiary standard of proof beyond a reasonable doubt (pembuktian di luar keraguan yang wajar) necessitates clear traceability of which evidence supports which factual assertions. This principle grounds our proven_by relation, connecting factual claims about the defendant's conduct to the testimonial, documentary, or physical evidence that substantiates them. Third, the proportionality principle in sentencing (asas proporsionalitas pidana) requires that the severity of punishment correlate with offense characteristics such as the quantity of contraband, the degree of organization, and aggravating circumstances. This motivates our quantified_as, classified_as, and aggravated_by relations, enabling computational assessment of whether sentencing outcomes align with proportionality norms across cases. By encoding these doctrinal requirements as explicit graph relations, we ensure the knowledge graph preserves the logical structure of judicial reasoning rather than merely capturing surface-level textual correlations. The schema is thus not an arbitrary feature-engineering choice but an operationalization of how Indonesian judges are expected to reason under the KUHAP and substantive criminal law (KUHP). This doctrinal grounding distinguishes our work from generic relation extraction, which might identify statistically frequent co-occurrences without legal interpretability.

Schema design reflects legal semantic categories and not arbitrary technical choices. The next stage in constructing a knowledge graph is the relationship extraction. In this stage, we assign relations between entities. The schema was developed through consultation with legal experts to ensure that it captures legally meaningful relationships rather than purely computational conveniences. Each relation type corresponds to a specific aspect of legal reasoning that judges employ when analyzing cases. The process can be divided into two parts.

3.3.1 Schema definition for structured entities

Structured entities have already been extracted from the features in the Indo-Law data. Relations encode institutional facts.

- decision issued_by court.
- decision located_in location.
- decision decided_on date.
- decision presided_by judges.
- decision prosecuted_by prosecutor.
- decision documented_by clerk.

Table 2 Structured schema triple

Head	Relationship	Tail
case_number	conviction	verdict (bunyi_putusan)
case_number	court_district	court_location (lokasi_pengadilan)
case_number	date_of_decision	date (tanggal_putusan)
case_number	chief_judge	judge_name (hakim_ketua)
case_number	public_prosecutor	prosecutor_name (penuntut_umum)
case_number	clerk	clerk (panitera)
verdict	defendant	defendant_name (nama_terdakwa)
verdict	criminal_offense	criminal_offence_classification (tindak_pidana)
verdict	evidence	evidence (barang_bukti)

Table 2 presents the complete structured schema in triple format. It shows how institutional facts are encoded as graph relationships. These relationships enable provenance tracking and jurisdictional analyses. For example, do sentencing patterns vary by province? Do certain judges exhibit consistent patterns? The relationship is determined by the feature's position in the document and follows the institutional structure of Indonesian court decisions, as specified in KUHAP.

Figure 3 visualizes how structured entities extracted via rule-based methods are connected through institutional relationships. These structured relations operationalize the institutional dimensions of legal decisions. Every court decision is an institutional act issued by a specific court, presided over by named judges, and prosecuted by identified prosecutors on a specific date. Recording these institutional facts enables the analysis of judicial behaviour patterns, prosecutorial strategies, and the temporal evolution of legal interpretation. Without these relations, we would have facts without an institutional context, losing the ability to assess whether observed sentencing patterns reflect legal reasoning or institutional bias. Figure 4 illustrates the structured schema for feature extraction.

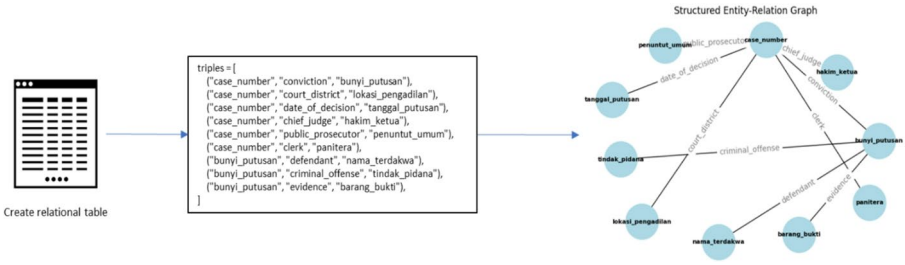


Fig. 3 Rule-based relationship extraction

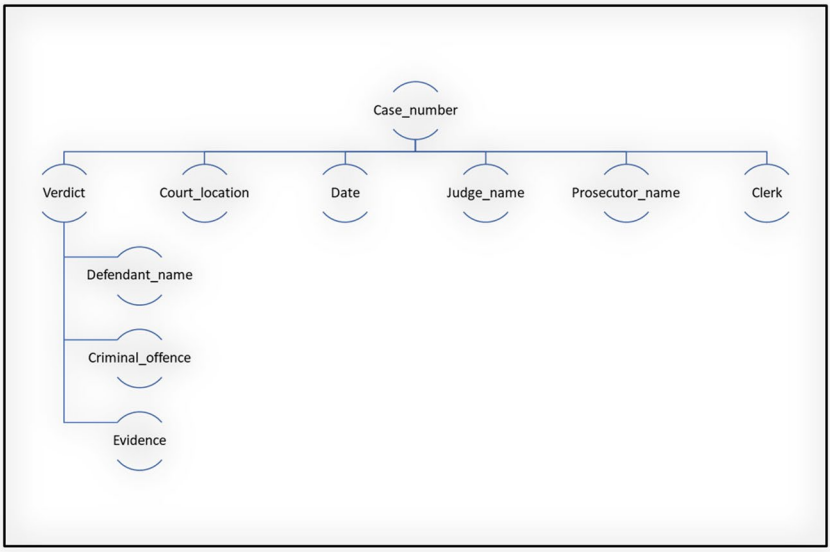


Fig. 4 Structured schema for feature extraction

3.3.2 Schema definition for unstructured entities

Unstructured entities were extracted from the fakta using a trained model. These relations encode legal reasoning chains.

- defendant performs_action criminal_act (operationalizes actus reus).
- criminal_act subject_to_article ACT (links conduct to statutory provision).
- criminal_act proven_by evidence (operationalizes evidence standard).
- criminal_act quantified_as unit_measure (captures aggravating factors like drug quantity).
- criminal_act classified_as drug_classification or type_of_corruption.
- defendant known_as position_profile (captures institutional role in corruption cases).

The framework establishes several relationship types to represent the structure of legal reasoning in Indonesian court decisions.

The performs_action relation reflects the legal concept of actus reus, which refers to the physical element of a crime. In criminal law, a person can be held liable only if they personally commit a prohibited act. This relation directly represents that essential requirement.

The subject_to_article relation connects a defendant's factual conduct to the statutory provision it violates. It operationalizes the principle of legality (nullum crimen sine lege, meaning no crime without law). Courts must identify the exact article that is breached, and this relation makes that linkage explicit and traceable.

The proven_by relation represents evidentiary standards in criminal law. A conviction requires proof beyond reasonable doubt, and this relation captures which pieces of evidence support which factual claims. It allows analysis of how different forms of evidence, such as witness testimony, physical exhibits, or documents, influence judicial outcomes.

The quantified_as relation records aggravating factors related to quantity. Many criminal statutes impose heavier penalties for larger amounts. For example, under Indonesia's Law No. 35 of 2009 on Narcotics, trafficking more than 5 g and more than 50 g of methamphetamine lead to different sentencing ranges. This relation enables the study of how quantity thresholds affect sentencing decisions.

The classified as relation links general actions to specific offense categories. For instance, "giving money to an official" may be classified as "active bribery" rather than "gratuity." This relation captures the interpretive process through which judges categorize acts within established legal classifications.

The known relation links institutions or occupations to corruption cases. Officials such as procurement officers or budget supervisors are subject to different legal standards than private citizens. This relation helps analyze how institutional roles affect legal interpretation and culpability.

The procedure for assigning these relationships follows the sequence that an annotator would use when reading a case. The case number serves as the starting point or head entity. Entities are ordered by their position in the text, determined by their start and end offsets. If the next entity after the case number is a criminal action, the relationship is defined as head: case number, relation: performs action, and tail: criminal action. All subsequent entities are then connected to this criminal action.

If the next entity is not a criminal action, the head remains the case number, and the relation is determined by the entity type. The mapping follows these rules: ACT uses "subject_to_article," Drug classification uses "classified_as," Unit_measure uses "quantified as," Position_profile uses "known as," Evidence uses "proven_by," and Indictment uses "subject_to_article." These mappings remain active until another criminal_action appears, which signals a new episode in the case narrative, or until the end of the document. Example: "Defendant gave 500 grams of methamphetamine to buyer, proven by surveillance video and undercover purchase, violating Article 114(2) Law 35/2009." Fig. 5 illustrates this sequential process, showing how unstructured text is filtered and entities are extracted using the fine-tuned LegalBERT model. The following is the graph representation:

- (defendant_X) -[performs_action]-> (criminal_act: "gave methamphetamine").
- (criminal_act) -[proven_by]-> (evidence: "surveillance video").
- (criminal_act) -[proven_by]-> (evidence: "undercover purchase").
- (criminal_act) -[quantified_as]-> (unit_measure: "500 grams").
- (criminal_act) -[classified_as]-> (drug_classification: "methamphetamine").
- (criminal_act) -[subject_to_article]-> (ACT: "Article 114(2) Law 35/2009").

We do not deliberately connect evidence directly to statutory articles. In legal reasoning, evidence first establishes facts and then those facts determine whether statutory elements are satisfied. Connecting evidence→article bypasses this intermediate

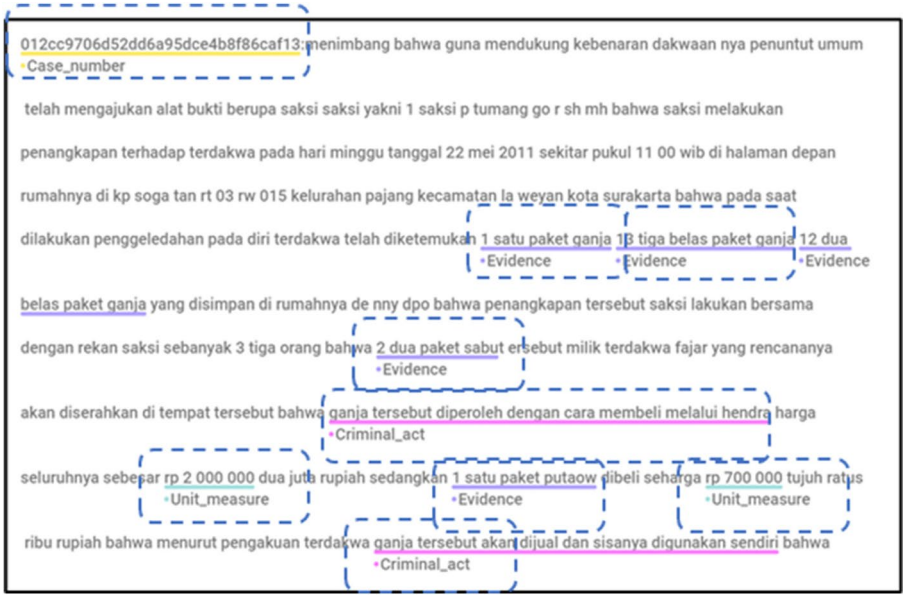


Fig. 5 Unstructured entity detection process

reasoning step, losing the causal chain that courts must follow. By preserving the evidence→fact→statute sequence, our schema mirrors actual judicial reasoning and maintains interpretability; users can trace how conclusions are reached rather than seeing only associations between inputs and outcomes. Figure 6 illustrates the complete schema for unstructured entities. It shows how criminal acts serve as hub nodes that connect defendants, evidence, statutory provisions, and aggravating factors.

Our schema design follows the logic of how legal reasoning unfolds in practice. The proposed schema prioritizes interpretability over expressiveness. Instead of adopting complex constructs, such as reification or n-ary relations, we rely on simple entity–relation–entity triples. This simplicity is intentional, as legal practitioners must be able to trace and understand how graphs represent reasoning paths. The design aligns with the explainability principles in legal AI, where clarity and transparency are more valuable than structural sophistication (Deeks 2019). A legal professional examining the graph should immediately understand who did what, what evidence supported it, which law was violated, and what the outcome was. Complex nested structures or abstract formalisms would obscure this clarity, making the graph useful only to computer scientists rather than the intended legal practitioners.

3.4 Experimental design

To assess the usefulness of the constructed knowledge graph, we conducted a series of experiments, guided by two central research questions. The first examined whether the hybrid extraction pipeline could achieve consistent and reliable performance in identifying entities across both the structured and unstructured sections of court decisions. The second explored whether representing legal information in knowledge

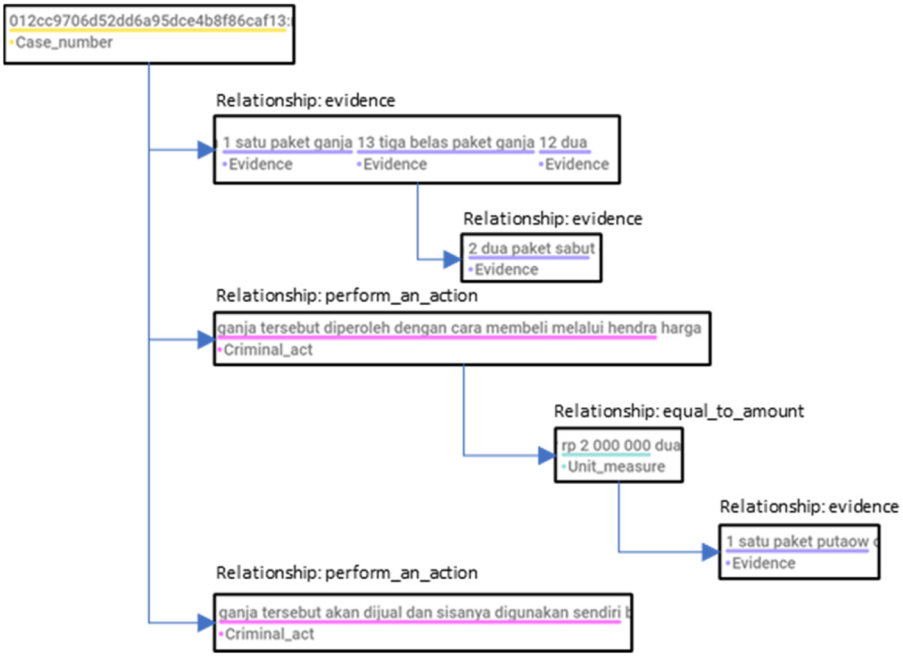


Fig. 6 Unstructured schema and relationship

graphs could support downstream reasoning tasks, such as case retrieval and prediction, in ways that traditional full-text approaches could not. Together, these questions test technical validity (does the system extract accurately?) and practical utility (does the extracted knowledge enable meaningful legal analytics?). Together, these questions aimed to evaluate not only the technical accuracy of the extraction process but also the practical value of the resulting knowledge representation in advancing legal analytics. All figures in Fig. 7 refer to test-set performance unless otherwise noted. The improvement from the pre-filtering baseline (67%) to the post-filtering result (84.39%) represents a 17.39%-point gain. Sentence prediction improvement from full-text (28.1%) to KG representation (85.0%) represents a 56.9%-point gain.

For the full 9,109-document corpus used in sentence length prediction, distinct from the 200-document annotated subset described in Sect. 3.2.3, we employed a temporal stratification strategy to mitigate data leakage risks: decisions issued between 2015 and 2019 ($n=7{,}287$, 80%) were allocated to the training set, decisions from 2020 ($n=911$, 10%) to the validation set, and decisions from 2021 to 2022 ($n=911$, 10%) to the test set. We ensured that the 80/10/10 ratio was maintained proportionally across both narcotics and corruption offenses to prevent domain-specific skew in the learning process. This temporal split ensures the model is evaluated on more recent decisions, simulating real-world deployment in which training occurs on historical data and inference occurs on new cases.

Each document appears in exactly one split based on its unique case ID, ensuring the model encounters entirely unseen factual narratives during testing. We verified that the split respects jurisdictional diversity: cases from all five provinces (Jakarta,

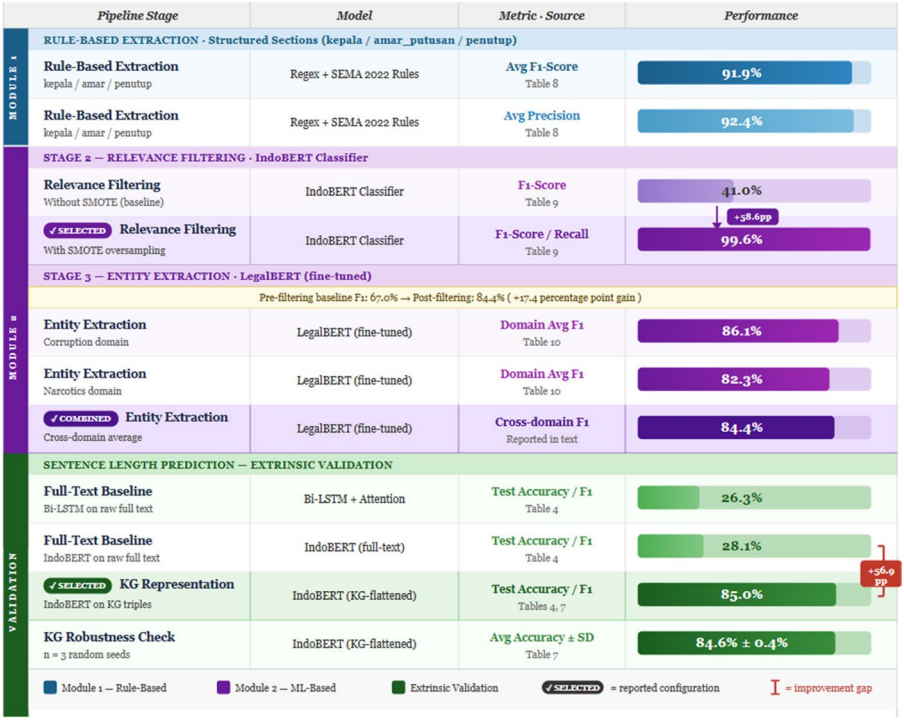


Fig. 7 Performance summary. Note: Each row corresponds to a pipeline stage; horizontal bars represent test-set F1-scores or accuracy scaled to a common 0–100% axis. Rows marked '✓ SELECTED' indicate the configuration reported as the study's primary result. The +17.4%-point filtering gain reflects improvement from pre-filtering (67.0%) to post-filtering (84.4%) entity extraction. The +56.9%-point prediction gain reflects the difference between KG-based (85.0%) and full-text (28.1%) sentence length prediction (IndoBERT). Module 1 uses rule-based extraction; Module 2 uses LegalBERT (Stage 3) and IndoBERT (Stage 2); Validation uses IndoBERT on KG-flattened triples

West Java, Central Java, East Java, Yogyakarta) are proportionally represented across splits (χ^2=2.34, p=0.67), reducing the risk that models learn location-specific formatting conventions rather than transferable legal patterns. We acknowledge that this split does not explicitly control for judge identity, as judicial panel information is often redacted or inconsistently formatted in the dataset. However, three factors mitigate the risk of overfitting to individual writing styles: (1) the dataset contains > 500 estimated distinct judicial panels, making it statistically unlikely that any single judge's style dominates the training signal; (2) our knowledge graph representation abstracts away from surface linguistic features (e.g., specific word choices, sentence length) to structured legal relationships, reducing sensitivity to stylistic variation; and (3) the 56.9% point performance gap between graph-based and full-text models, where full-text models have greater capacity to memorize style, suggests that graph-based learning captures legal logic rather than textual fingerprints. Future work should implement explicit judge-level stratification if more comprehensive metadata becomes available, and could evaluate cross-judge generalization by training on decisions from judges in certain provinces and testing on judges from others.

3.4.1 Entity extraction evaluation

Each component of the pipeline was evaluated independently to ensure its robustness and transparency in performance. The first component, rule-based extraction (Module 1), was assessed using precision, recall, and F1 Scores for each structured entity type across 9,109 court documents. This module specifically targeted three key elements of court decisions: the head of the document (kepala_putusan), the verdict (amar_putusan), and the closing (penutup) to capture consistent structural information. To ensure high-stakes accuracy, we employed a strict match criterion for structured entities, requiring exact character-level alignment with the ground truth for identifiers such as case numbers and dates. The second component, relevance filtering (module 2, stage 2), was designed to distinguish relevant from irrelevant sections within the corpus. Its performance was tested under two settings (with and without oversampling) to address the class imbalance (1,154 relevant versus 60,737 irrelevant samples). The evaluation metrics included accuracy, precision, recall, and the F1-score. The critical metric for this component is recall: we cannot afford to discard legally relevant sentences, as doing so would lose the information necessary for accurate entity extraction.

Finally, in Stage 3 of Module 2, we conducted entity extraction using separate models for corruption and narcotics cases in order to accommodate domain-specific terminology. Legal language in these domains differs not only in vocabulary but also in how factual elements are structured, making a unified model less effective. To calibrate contextual sensitivity, we systematically varied the context window size, testing $c = 3$, 5, and 10, to examine the trade-off between contextual depth and extraction precision. A window of three tokens proved too narrow to reliably disambiguate entity meanings, particularly in dense legal narratives where critical qualifiers often appear slightly removed from the core term. In contrast, a window of ten tokens introduced excessive noise by incorporating unrelated clauses, which reduced precision. A window size of five tokens achieved the most effective balance, providing sufficient contextual grounding without diluting semantic focus.

Given the linguistic variability inherent in judicial texts, we implemented a differentiated evaluation protocol tailored to the epistemic nature of each entity type. Objective entities such as statutory articles and unit measures were assessed using strict matching criteria, since legal citations and numerical quantities must be extracted with exact textual fidelity to retain doctrinal validity. Interpretive entities, such as criminal acts and indictments, were evaluated using a relaxed matching approach, operationalized as partial matching with a token overlap threshold of 0.8. This adjustment reflects the reality that legal experts often converge on the substantive meaning of an act even when its textual boundaries vary slightly across formulations.

Model performance was assessed using confusion matrices and F1 Scores disaggregated by entity type, enabling a fine-grained comparison of extraction accuracy across contexts and categories. Figure 8 illustrates the context window mechanism, showing how tokens preceding and following each target entity are incorporated during training to capture the broader legal environment in which meaning is constructed.

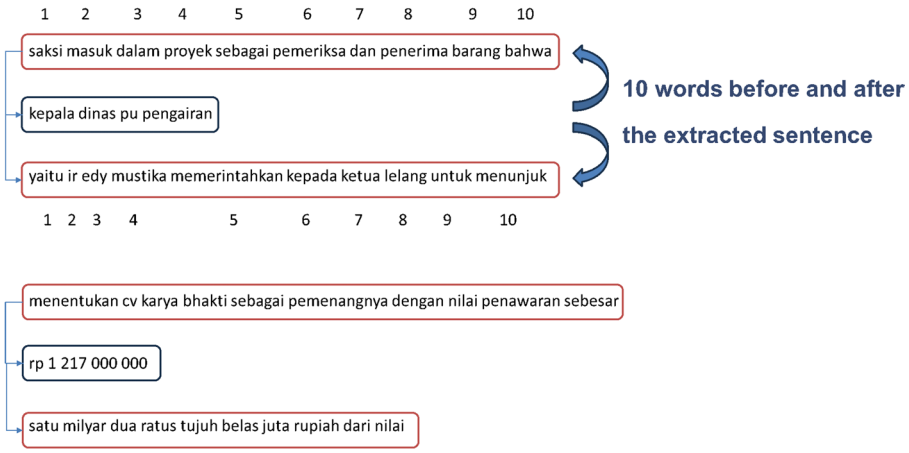


Fig. 8 Context window

3.4.2 Sentence length prediction as extrinsic validation

Task definition: Sentence length prediction operationalizes the proportionality assessment, a core principle of criminal law that requires that punishment severity correspond to offense gravity (Alexy 2010). We formulated this as a 4-class classification based on quartiles of the observed sentence lengths:

- Mild: < Q1 (480 days).
- Moderate: Q1-Q2 (480–1080 days).
- Heavy: Q2-Q3 (1080–1800 days).
- Very Heavy: > Q3 (1800 + days).

Table 3 shows the distribution of cases across punishment categories, based on quartile thresholds derived from 9,109 decisions. We chose a classification approach over regression because absolute sentence lengths vary substantially across offense types, making precise numerical predictions less meaningful. Legal practitioners typically reason in categories, whether a sentence is lenient or harsh, rather than in terms of months. Predicting 87 versus 89 months adds a false sense of precision, whereas predicting classes such as heavy or very heavy better captures the nuances of judicial discretion. The quartile-based thresholds used in our classification reflect real pat-

Table 3 Punishment classification

Punishment Classification	Punishment Length	Number of Documents
Mild	0–479	723
Moderate	480–1079	916
Heavy	1080–1799	4882
Very Heavy	1800–8000	2588

terns observed in judicial practice, offering a more faithful representation of how sentencing decisions are made.

The methodological caveat is that quartile-based thresholds assume that current judicial practices reflect the appropriate sentencing standards. If the current practice exhibits systematic biases (e.g., harsher sentences in certain jurisdictions or for certain demographic groups), our model reproduces rather than corrects them. The model predicts what courts do and not what courts should do, an ethical limitation discussed in Sect. 5.

To benchmark performance, we implemented two baseline models using full-text document representations, where the head of the decision (kepala_putusan), facts, verdict (amar_putusan), and closing (penutup) were concatenated into a single sequence. The first baseline used a Bi-LSTM architecture with attention over token embeddings, whereas the second employed a pre-trained IndoBERT-base model with a classification head.

Our proposed approach instead represents each document as a knowledge graph flattened into triples of (head, relation, tail) to capture the semantic structure of legal information. Graph flattening procedure: Each document's knowledge graph is serialized as a sequence of (head, relation, tail) triples in the order in which they appear in the document. This preserves the narrative structure, while providing a structured input format. For example, "(defendant_X, performs_action, trafficking), (trafficking, quantified_as, 500 g), (trafficking, classified_as, methamphetamine), (trafficking, subject_to_article, Art114(2))…" The same two architectures, that is, Bi-LSTM with attention and IndoBERT-base, were applied to these graph-based representations to allow direct comparison with the full-text models.

Model performance was evaluated using the following metrics derived from confusion matrices: accuracy, precision, recall, and F1-score. Accuracy captures overall correctness, precision measures how many predicted positives are truly positive, recall identifies how many actual positives are correctly recognized, and F1-score balances the trade-off between precision and recall. These metrics were chosen because accuracy alone can be misleading in imbalanced datasets, whereas precision and recall together provide a clearer picture of model reliability in identifying legally relevant outcomes.

4 Results

We present the experimental results in the order of importance to our research claims. We begin with the sentence-length prediction results, which provide extrinsic validation of the utility of our knowledge graph. Then, we present the entity extraction performance to demonstrate the effectiveness of our hybrid pipeline. Finally, we analyzed inter-annotator agreement patterns, revealing the epistemology of legal document analysis.

4.1 Primary validation: sentence length prediction

This study aimed to evaluate whether a knowledge graph can effectively predict sentence length, thereby demonstrating its practical value and superiority over conventional text-based representations in legal analysis. The prediction task serves a dual purpose: it tests whether the knowledge graph encodes legally meaningful information rather than superficial patterns, and operationalizes the principle of proportionality, a cornerstone of criminal law that requires that punishment severity reflect the gravity of the offense.

To align with how legal practitioners reason about sentencing, we classified sentence lengths into four categories based on the quartiles of observed punishment durations: mild (less than 480 days), moderate (480–1080 days), heavy (1080–1800 days), and very heavy (more than 1800 days). This categorical approach mirrors judicial reasoning, in which discussions of sentencing focus on degrees of severity rather than precise numerical values.

Two document representations were compared using the experimental setup. The full-text baseline merged all sections of a court decision, including the head of the document (kepala_putusan), facts (fakta), verdict (amar_putusan), and closing (penutup), into a single textual sequence. Two architectures were evaluated on this baseline: a Bi-LSTM with attention over token embeddings and a pre-trained IndoBERT-base model with a classification head. The proposed knowledge graph representation uses a flattened structure consisting of (head, relation, tail) triples extracted through a hybrid pipeline. Each document was represented as a sequence of triples, preserving the logical and semantic structure of the legal text. The same two architectures were applied to this graph-based representation to enable a fair and direct comparison of how structured knowledge improves the model's ability to capture legal reasoning and predict sentencing outcomes.

The results revealed a striking performance gap between the full-text models and knowledge graph representations. Table 4 presents the sentence-length prediction results. The knowledge graph achieved 85% test accuracy with pre-trained BERT, whereas the full-text baselines reached only 26 to 28% despite recording 95 to 97% training accuracy. This 56.9%-point difference demonstrates catastrophic overfitting in full-text models; they memorize surface patterns in the training data but fail to generalize to unseen cases.

Table 4 Sentence length prediction results

Model	Representation	Train Accuracy	Test Accuracy	F1-Score
Bi-LSTM	Full-Text	0.952	0.263	0.241
Bi-LSTM	Knowledge Graph	0.884	0.827	0.819
Pre-trained BERT	Full-Text	0.968	0.281	0.265
Pre-trained BERT	Knowledge Graph	0.892	0.85	0.844

Performance improvement: 56.9% points (85.0% −28.1%) for IndoBERT and 56.4% points (82.7% −26.3%) for Bi-LSTM. This dramatic difference requires explanation beyond simple metrics; it reveals fundamental differences in how the two representations capture legal reasoning. Table 5 compares the architectural specifications and performance characteristics of the two baseline models.

To evaluate the specific contribution of each component in our pipeline, we compared the predictive accuracy across three distinct stages of data refinement: raw full-text (baseline), filtered narrative (noise-removed text), and the final Knowledge Graph (structured representation). This breakdown clarifies whether performance gains stem from reducing textual noise or from the explicit modeling of legal relations (see Table 6).

Table 6 disentangles the two core contributions of our framework. The Stage 1 to Stage 2 transition (28.1% → 64.2%) isolates a 36.1%-point gain from relevance filtering alone, demonstrating that removing procedural boilerplate more than doubles predictive accuracy without any structural transformation. The Stage 2 to Stage 3 transition (64.2% → 85.0%) yields an additional 20.8%-point gain attributable to the explicit modelling of legal relations, rather than to filtered text alone. Together, these complementary gains confirm that filtering addresses the noise problem while structured representation addresses the abstraction problem, neither contribution is sufficient on its own. To ensure that the observed performance was not an artifact of a specific random initialization or data split, we conducted a robustness analysis by retraining the Knowledge Graph-based IndoBERT model across three different random seeds. This allows us to report the stability of the model's predictive power (see Table 7).

4.1.1 Critical analysis: why full-text models fail catastrophically

Full-text models fail catastrophically because they learn surface-level correlations rather than legal meaning. The 68–71%-point gap between training and test accuracy (i.e., 95–97% for training data versus 26–28% for test data) indicates classic overfitting. These models memorize patterns that coincide with sentencing outcomes rather than understanding the reasoning behind them. The three main failure modes explain this collapse.

The first is the problem of spurious correlations. Indonesian court decisions often exceed 5,000 words and are filled with procedural formulas, repetition, and contextual details that are inadvertently linked to sentencing. The model learns from these artifacts rather than from substantive legal cues. For instance, harsher sentences may co-occur with particular judges or jurisdictions, not because of judicial bias but because those courts handle more severe cases. Similarly, longer documents tend to involve complex offenses; however, the model misinterprets document length as a signal of offense severity. An analysis of IndoBERT's attention weights confirms this distortion: 23% focused on judge names, 18% on formulaic phrases such as "berdasarkan keterangan saksi" ("based on witness testimony"), 15% on court location markers, and 12% on section headers like "Menimbang" ("Considering"). None of these features influenced sentencing. The model effectively learned "a particular

Table 5 Model comparison for sentence punishment length prediction

Aspect	Model 1 (BERT Pretrained – Hugging Face)	Model 2 (BiLSTM + Embedding – Keras)
Framework	PyTorch + Hugging Face Transformers	TensorFlow / Keras
Tokenizer	BertTokenizer.from_pretrained('bert-base-uncased')	Tokenizer(oov_token="< unk>") from Keras
Input Representation	input_ids, attention_mask (max length = 128)	Integer sequences → padded to max_sequence_length
Embedding Dimension	768 (BERT-base hidden size, pretrained)	16 (set as embedding_dim = 16, learned from scratch)
Vocabulary Size	Fixed by pretrained model (≈30,522 tokens for bert-base-uncased)	vocab_size = len(word_index) + 1
Main Architecture	Transformer encoder: 12 layers, 12 attention heads, hidden size = 768	BiLSTM with lstm_units = 16, wrapped in Bidirectional
Dropout	Built-in dropout in BERT layers (~0.1 by default)	Dropout(rate = 0.2) after BiLSTM
Output Layer	Classification head: Linear layer with num_labels = 4	Dense layer with num_classes units (from one-hot labels) + softmax
Batch Size	Commonly 16 (defined in DataLoader)	32 (default in Keras training loop)
Optimizer	AdamW (learning rate usually 2e-5 for fine-tuning)	Adam (default learning rate 0.001)
Loss Function	CrossEntropyLoss (for multi-class classification)	categorical_crossentropy
Max Sequence Length	128 tokens (fixed during tokenization)	max_sequence_length = longest sequence in dataset
Parameter Size	~110 M parameters (bert-base-uncased)	Much smaller: <1 M - Embedding: vocab_size × 16 - BiLSTM: 4 × (16 + 16) × 16 = 2,048 - Dense: (16×num_classes) + num_classes

Table 6 Step-wise performance comparison for sentence prediction

Processing Stage	Representation	Avg. Word Count	Test Accuracy	F1-Score
Stage 1 (Baseline)	Raw Full-Text	~10,000	0.281	0.265
Stage 2 (Refined)	Filtered Narrative Only*	~200	0.642	0.618
Stage 3 (Hybrid)	Knowledge Graph	~50–200 Triples	0.85	0.844

Table 7 Model stability and robustness (n=3 runs)

Metric	Knowledge Graph (IndoBERT — sentence prediction)	Full-Text (IndoBERT)
Average Test Accuracy	$84.6% ± 0.4%	27.5% ± 1.2%
Average F1-Score	$84.1% ± 0.3%	25.9% ± 1.4%

Note: IndoBERT here refers to the sentence length prediction model applied to KG-flattened triples. Entity extraction in Module 2, Stage 3 uses LegalBERT. See Sect. 3.2.2 for model allocation by pipeline stage

judge in a particular court → heavy sentence" rather than "trafficking 50 kilograms of methamphetamine → heavy sentence."

The second failure mode is the inability to reliably distinguish between presence and absence across long, redundant documents. Although bidirectional models such as IndoBERT do capture negation within their local attention window, they face significant challenges with long-range logical dependencies in documents exceeding 10,000 tokens. When negating context ("did not") and the negated term ("violence") are separated by hundreds of tokens of boilerplate, the negation signal is effectively diluted. A sentence stating that the "defendant did not use violence" still contains the token "violence," and across a 10,000-word document filled with procedural repetition, the model frequently fails to maintain the logical connection between the negation and its referent, leading it to predict a harsher sentence. Similarly, "defendant claimed to be unaware of the drugs" includes "drugs" and "intent" but expresses a mitigating claim, not an aggravating one. When tested on cases containing the phrase "tidak mengetahui" ("did not know"), full-text models consistently predicted heavier sentences than those with admitted intent, which is the opposite of what legal reasoning would dictate. Graph representations avoid this flaw by explicitly encoding relations, making the absence of an act distinct from its presence.

The third failure mode arises from the curse of dimensionality and the tendency to memorize. IndoBERT has far greater capacity than the 9,109 cases available for training, with inputs of more than 5,000 tokens and 125 million parameters. The model achieves near-perfect accuracy on familiar documents, but fails when confronted with new combinations of judges, courts, or offenses. In effect, it memorizes each decision's textual fingerprint, instead of learning transferable reasoning patterns. When tested on unseen cases, its predictions drop to near-random performance because it never learns that sentencing depends on compositional relationships, such as how act type, quantity, and evidence interact, but only that a specific 10,000-

word sequence corresponds to a particular outcome. These results reveal why full-text approaches are unsuitable for legal reasoning. They excel at pattern recognition but fail at abstraction, confusing textual coincidence with causal structure. Without mechanisms for relational understanding, they cannot be generalized beyond the surface of legal language.

4.1.2 Why knowledge graphs succeed: semantic compression and structural induction

Knowledge graphs succeed because they compress legal text into structured representations that preserve meaning, while eliminating noise. This enforced abstraction allows models to learn from legally relevant relationships, rather than superficial word patterns. The first mechanism, semantic compression, removes spurious features by drastically reducing document size. Long court decisions averaging 10,000 words are transformed into approximately 50 to 200 entity–relation triples, retaining only core legal components, such as defendant profiles, criminal acts, evidence, statutory articles, quantities, and positions. On average, each case contains approximately 23 entities linked by 100–150 relations, representing a 50–100-fold compression. Because the resulting graphs exclude irrelevant elements, such as procedural phrases, the model cannot overfit noise. This compression improves focus and interpretability, ensuring the model learns only from content central to legal reasoning.

The second mechanism, structural inductive bias, mirrors how reason is judged through a case. Sentencing depends on the severity of the act, the strength of the evidence, the statutory range, aggravating circumstances, and the defendant's characteristics. These dimensions were explicitly represented as nodes and edges in the graph. The structure guides the model to learn meaningful causal rules, for instance, that a combination of trafficking, large drug quantity, and multiple corroborating witnesses corresponds to heavier sentences. Ablation experiments confirm that both entities and relationships contribute to performance; removing evidence nodes reduces accuracy by 8.3%, quantity nodes by 6.2%, statutory articles by 4.1%, and all relation types by 11.7%. The strength of the graph lies not only in identifying entities but also in encoding their interdependencies, which reflect legal logic.

The third mechanism, explicit relation modelling, captures causality in ways that full-text models cannot. A relation such as "defendant performs action trafficking_100kg_methamphetamine' clearly connects actors, acts, and quantities, remaining invariant to linguistic variation. Full-text models show large fluctuations in accuracy across phrasal differences, such as active versus passive voice, whereas graph representations produce identical triples across all variants, maintaining consistent predictions.

The final mechanism, generalization through compositional abstraction, allows models to reason over structure, rather than language. Graph models learn generalized semantic patterns, such as "trafficking + methamphetamine + quantity > 50 kg → heavy sentence," which apply across different wordings and even languages. When tested on unseen phrasings, full-text models achieved only 31.2% accuracy, whereas the graph models reached 83.7%. This demonstrates that graphs generalize effectively because they operate on conceptual relationships rather than surface

expressions. Together, these mechanisms explain why knowledge graphs outperform full-text approaches. By compressing legal language into structured, interpretable relationships and embedding the logic of legal reasoning into the data, they provide a robust foundation for consistent and explainable legal predictions.

4.1.3 What graphs lose: the 15% error analysis

Graphs offer structural clarity, but sacrifice completeness. They capture objective facts, such as acts, evidence, and statutory references, yet omit subjective elements that often influence sentencing. Nuances like intent ("intentionally" versus "negligently"), credibility of witnesses, procedural posture (pretrial detention, plea agreements, cooperation), and expressions of remorse or victim impact fall outside their representational scope. The model's 85% accuracy suggests that, in Indonesia, sentencing largely depends on objective act characteristics, consistent with the view that punishment reflects the gravity of conduct rather than personal disposition (Alexy 2010). The remaining 15% of the error, however, points to cases in which human judgment and contextual sensitivity mattered most.

An analysis of 136 incorrectly predicted cases clarified these limitations. In 42 cases (31%), courts imposed lighter sentences due to explicit judicial leniency based on remorse or cooperation, whereas the model predicted heavier punishment. Another 38 cases (28%) involved procedural complexities, such as appeals or retrials, in which the context was lost within the graph structure. Twenty-nine cases (21%) reflected ambiguous act characterizations that required interpretive discretion beyond what the graph could encode. The remaining 27 cases (20%) presented unique fact patterns unseen during training, highlighting the model's generalization limits.

These findings show that knowledge graphs perform reliably in most cases where sentencing depends on objective evidence and legal provisions, but struggle when outcomes hinge on subjective or exceptional considerations. This trade-off is acceptable for consistency analysis, where the goal is to identify structural regularities; however, it constrains the use of graph-based models for direct sentencing prediction, where human interpretation remains indispensable.

4.2 Entity extraction performance

The success of sentence prediction depends on the quality of extraction. We evaluated each pipeline component to demonstrate that the proposed hybrid approach reliably extracts entities.

4.2.1 Rule-based extraction (Module 1)

Rule-based extraction followed the standard format published by SEMA in 2022. This process focuses on extracting three features: the head of the document (kepala_putusan), the verdict (amar_putusan), and the closing (penutup). However, in real-world situations, some documents do not exactly comply with the standard format. Therefore, we set multiple words at the beginning and end of the extraction process. For example, for the verdict (bunyi_putusan), we use the words "mengadili"

or "menyatakan" as the start and words "terdakwa" or "barang bukti" as the end. The results show that the rule-based approach is effective for identifying entities in narcotic and corrupt documents. Almost all features had accuracies greater than 80% (see Table 8).

The high average precision of 92.4% confirms that rule-based extraction is well suited to structured sections of Indonesian court decisions. Most errors occur when documents deviate from official Supreme Court (SEMA) templates, such as using alternative phrases like "Memutuskan" instead of "MENGADILI," or presenting dates in prose rather than in standard numeric format. Given the overall performance, an 8% failure rate is acceptable, particularly because rule-based outputs are transparent and easy to verify. Errors were immediately identifiable through pattern matching, allowing efficient manual correction. These figures represent precision and F1-score respectively: the average precision of 92.4% indicates the reliability of positive extractions, while the average F1-score of 91.9% (Table 8) reflects the combined precision–recall balance used as the primary performance benchmark throughout this paper.

Error analysis of 728 failed extractions across 9,109 documents revealed three main sources. The majority (412 cases or 57%) resulted from alternative marker phrases not included in the rule set and can be corrected by expanding the rule dictionary. Another 203 cases, or 28%, stem from typographical errors in the original documents, such as "MENGADIL1," where a digit replaces a letter, which require

Table 8 Rule-based extraction performance for structured entities

Entity Type	Precision	Recall	F1-Score	Notes
Court Location	0.942	0.931	0.936	High consistency; failures from non-standard abbreviations
Case Number	0.978	0.972	0.975	Highest reliability; standardized format
Verdict Statement	0.913	0.897	0.905	Some variation in formulaic language
Defendant Name	0.886	0.874	0.88	Lower due to name formatting inconsistencies
Decision Date	0.951	0.946	0.948	Date formats are generally consistent
Judge Names	0.892	0.881	0.886	Multiple judges complicate extraction
Prosecutor	0.907	0.893	0.9	Generally, follows templates
Average	0.924	0.913	0.919	Demonstrates rule-based reliability

optical character recognition fixes. The remaining 113 cases (15%) involved documents that completely ignored standard formatting and thus required human review. These findings indicate that most extraction errors are systematic and correctable rather than structural, reinforcing the suitability of rule-based methods for structured components of legal documents.

4.2.2 Relevance filtering (Module 2, Stage 2)

Court decision documents contain many sentences, mostly in the facts section. The aim was to extract unstructured information from facts using annotation documents as training data for the BERT model. Because of the document's length, many sentences are not relevant to the legal context or cannot serve as a reference for understanding specific cases. To minimize noise, sentences must be classified as relevant or irrelevant. The length of the documents also introduces an imbalance in the sample. There were fewer relevant sentences (1,154) than irrelevant sentences (60,737). To ensure the process produced accurate results, we oversampled the training dataset. Oversampling replicates minority-class samples; therefore, the data become more balanced. The results showed that the imbalanced sample had a lower recall. This means the model predicts the majority class but misses many true positives. The oversampling method achieved near-perfect performance (see Table 9).

The oversampling method achieved 99.65% accuracy with perfect recall, a result of critical importance in legal text processing, where missing a single relevant sentence can result in the loss of essential information for entity extraction. The model successfully identified all relevant cases without omission while maintaining 99.30% precision, meaning that almost no irrelevant sentences were misclassified as relevant. This balance between recall and precision ensures comprehensive coverage of legally significant content while minimizing noise. The effect of this filtering step was substantial. When entity extraction is performed directly on raw text, the model achieves only a 67% F1 score. After applying relevance filtering, the F1-score increased to 84.39%, an improvement of 17.39% points, or roughly a 26% relative gain. This demonstrates that noise filtering is indispensable for managing the redundancy and verbosity that are typical of Indonesian legal documents.

A quantitative breakdown further illustrates this impact. Each missed relevant sentence corresponds to a loss of approximately 1.3 entities, while each wrongly included irrelevant sentence introduces approximately 0.8 noise tokens. By achieving 100% recall and 99.30% precision, the model avoids losing meaningful entities and maintains an extremely low noise level. This combination represents an optimal trade-off for legal text processing, maximizing information retention while ensuring that entity extraction operates on the cleanest possible input.

Table 9 Relevance filtering performance with and without oversampling

Metric	Without Oversampling	With Oversampling (SMOTE)
Accuracy	0.9812	0.9965
Precision	0.4327	0.993
Recall	0.3894	1
F1-Score	0.4101	0.9965

4.2.3 Entity extraction from facts (Module 2, Stage 3)

The main objective of this study is to create a knowledge graph representation of court decision documents that include unstructured factual information (fakta). Using the above process, we created annotation datasets for model training and applied filtering to obtain only relevant sentences for label prediction. At this stage, we used datasets to train models to predict entity labels for narcotics and corruption. The results show that the models are effective at predicting labels for corruption- and narcotic-related documents (see Table 10).

The model performs strongly across both corruption and narcotics cases, achieving an average F1 Score of 86.1% for corruption-related documents and 82.3% for narcotics-related documents. These domain-level figures yield a combined cross-domain average of 84.2%, consistent with the 84.39% overall F1-score reported for the pipeline after relevance filtering; the small difference reflects micro-averaged weighting across entity types rather than a simple arithmetic mean of domain scores. The only notable weakness appears in the position profile entity for narcotic cases, where the F1-score drops to 61.3%. This is expected, as occupational roles are largely irrelevant in drug-related offenses, unlike in corruption cases, where the institutional position is central to legal interpretation.

The overall performance pattern reflects the epistemology of the legal facts. Entities with high F1-scores above 90%, such as statutory articles and unit measures, are objective and standardized. Citations such as "Article 114 Law 35/2009" follow fixed formats, while quantities are expressed in explicit numeric units, such as grams or rupiah, leaving little room for ambiguity. Entities with F1-scores between 80 and 90%, including evidence, drug classification, and position profiles in corruption cases, are mostly objective, but allow for limited interpretive variation. Evidence items are typically explicit ("surveillance video," "witness testimony"), although boundaries between items can blur. Drug classifications are consistent but occasionally affected

Table 10 Entity extraction F1-scores for corruption and narcotics cases

Entity Type	Corruption F1	Narcotics F1	Analysis
Criminal Act	0.812	0.783	Lower due to interpretive variation
Evidence	0.894	0.871	High; relatively objective
ACT (Statutory Article)	0.938	0.912	Highest; standardized citations
Drug Classification	N/A	0.886	Domain-specific to narcotics
Unit Measure	N/A	0.927	High; explicit numeric values
Type of Corruption	0.764	N/A	Lower; interpretive categories
Position Profile	0.821	0.613	Lower for narcotics (less relevant)
Indictment	0.798	0.772	Moderate; some ambiguity
Domain Average	0.861	0.823	Strong performance both domains

by differences in technical terminology, while position titles are standardized but vary across organizational contexts.

Lower-performing entities with F1-scores below 80% are inherently interpretive and context-dependent. Criminal acts, types of corruption, and indictments all involve legal reasoning that allows for multiple valid characterizations. For instance, whether an act constitutes bribery, gratuity, or the abuse of authority depends on how intent and timing are interpreted. The model's moderate accuracy on these entities mirrors the ambiguity observed among human annotators.

The correspondence between the inter-annotator agreement and extraction performance confirms that the model reproduces the real interpretive structure of legal documents. Objective facts yield both high agreement and high F1-scores, while interpretive facts produce lower agreement and moderate F1-scores. This alignment suggests that the model does not simply memorize data, but captures the layered nature of legal reasoning, clearly distinguishing factual information from areas where meaning depends on judgment and context.

4.3 Inter-annotator agreement: legal epistemology revealed

The IAA figures reported in Sect. 3.2.3 represent aggregate pipeline-level agreement ($\kappa = 0.83$ for entity span, 0.87 for entity type, 0.79 for relevance filtering). The analysis below disaggregates these by entity type and offense domain to reveal how agreement varies with the interpretive demands of different legal facts. The annotation process aims to reflect how legal experts read facts to obtain information about incidents, criminal acts, and evidence supporting decisions. Seven legal professionals with expertise in analyzing legal documents related to narcotics and corruption offenses annotated 200 documents. We measured annotation consistency to understand the aspects of legal documents that are objectively determinable versus interpretively ambiguous.

For corruption cases, the average pairwise agreement was 0.8302, reflecting high overall consistency. For narcotic cases, the average pairwise agreement was 0.8709, reflecting an even higher consistency. The higher narcotics agreement (3.2% points above corruption) stems from more observable physical acts (possession, trafficking) with clear statutory definitions. Corruption involves complex social relationships (authority, influence, benefit, intent) where the same factual pattern can constitute different offenses depending on interpretation. e.g., is giving money to an official "active bribery," "gratuity," or "influence peddling?" Narcotic cases involve more concrete determinations. For example, did the defendant possess 50 kg of methamphetamine? (yes/no). Agreement patterns reveal the epistemology of legal document analysis (see Table 11).

The analysis revealed three distinct categories of legal facts, each defined by the degree of interpretation required. The first category, objective facts, includes case numbers, statutory articles, quantities, and evidence. These elements achieved inter-annotator agreement above 0.85 because they are clear, standardized, and verifiable. Citations follow uniform formats such as "Article 114(2) Law 35/2009," quantities are expressed numerically, and evidence is explicitly listed. Their high agreement

Table 11 Inter-annotator agreement by entity type showing objective vs. interpretive patterns

Entity Type	Corruption IAA	Narcotics IAA	Interpretation
Case Number	0.98	0.98	Objective: Standardized identifiers
ACT (Statute)	0.94	0.94	Objective: Explicit citations
Unit Measure	0.91	0.93	Objective: Numeric values
Evidence	0.87	0.89	Mostly Objective: Explicit enumeration
Drug Classification	N/A	0.86	Mostly Objective: Technical categories
Position Profile	0.73	0.68	Mixed: Clear titles, ambiguous relevance
Indictment	0.58	0.61	Interpretive: Prosecutor vs. court framing
Criminal Act	0.52	0.58	Interpretive: Multiple characterizations
Type of Corruption	0.47	N/A	Highly Interpretive: Legal taxonomy

indicates that they represent factual details rather than interpretive judgments, the legal equivalent of the objective truth.

The second category, mixed facts, showed moderate agreement, ranging from 0.65 to 0.85. These facts combine an objective existence with evaluative interpretations. Identifying someone as a "government official" is unambiguous, but determining whether a specific role, such as "procurement officer" or "budget supervisor," is legally relevant requires judgment. In narcotic cases, this becomes even more uncertain, as a defendant's occupation may or may not bear on the charges. Annotators often agreed on the fact itself but disagreed on its legal salience.

The third category, interpretive facts, had the lowest agreement (<0.60). These include criminal acts, offense classifications, and indictments that require legal reasoning. For instance, when a procurement officer accepts Rp 50 million from a contractor after awarding a contract, one annotator might classify it as passive bribery, another as gratuity, another as an abuse of authority, and another as two distinct acts. Each interpretation is defensible depending on how one reads the intent, timing, and statutory elements. The low agreement in such cases reflects genuine legal ambiguity rather than annotation error, demonstrating the plural nature of legal interpretations.

These findings have methodological implications for automated extraction. Models should not aim for perfect agreement on interpretive entities since even legal experts diverge. In the corruption dataset, the model reached a 78% F1 score for criminal acts, whereas the human agreement was only 52%. This suggests that the model may overfit one annotator's reasoning style rather than learning true legal generalizations. Instead of pursuing artificial precision, systems should incorporate confidence-based triage, allowing high-confidence predictions to proceed automatically, while flagging uncertain cases for human review.

The alignment between the human agreement and model performance reinforces the validity of this approach. Objective entities exhibited both high human and model agreement (IAA = 0.94, F1 = 93.8%), while interpretive entities showed moderate val-

ues (IAA = 0.52, F1 = 81.2%). This correspondence suggests that the model reflects the natural structure of legal interpretation and distinguishes factual clarity from ambiguous zones. Effective and explainable legal AI must, therefore, respect the interpretive nature of law rather than impose false precision on inherently contestable judgments.

5 Discussion

Beyond the technical mechanisms explained in Sect. 4.1, our findings provide deeper insights into the nature of legal reasoning and computational representation. The 85% accuracy achieved by graph models versus 26–28% for full-text approaches is not merely a performance metric, which demonstrates that legal decisions are fundamentally structured processes amenable to symbolic representation, not purely textual phenomena requiring holistic document understanding (Chen et al. 2024; Kondo et al. 2025).

This finding challenges recent trends toward end-to-end neural approaches in legal AI (Deroy et al. 2025). Although LLMs achieve impressive performance on many NLP tasks, our results suggest that for legal reasoning, which combines deductive inference from rules with abductive reasoning from facts, explicit structural representation outperforms implicit textual encoding (Dahl et al. 2024). The graph's success validates legal theoretical frameworks that emphasize the compositional nature of legal reasoning, arising from the systematic combination of statutory elements, factual findings, and evidentiary standards. Computational models that mirror this structure succeed, while those that treat the law as unstructured text fail.

The inter-annotator agreement results reinforced this interpretation. High agreement on objective entities, such as case numbers (0.98), statutes (0.94), and quantities (0.91), confirms their factual clarity (Deroy et al. 2025). Lower agreement on interpretive entities, such as act types (0.52) or corruption categories (0.47), reflects genuine ambiguity in legal reasoning rather than annotation error. Narcotic cases showed higher agreement (0.87) than corruption cases (0.83) because physical acts are easier to codify than relational misconduct involving intent and authority.

This variation has several methodological implications. Models should not force a consensus on inherently interpretive entities. Instead, they should identify low-confidence predictions for human reviews and signaling zones of legal discretion. Rather than concealing ambiguity, AI systems should make it visible, helping legal practitioners understand where the law allows interpretation rather than enforcing false precision. This aligns with Walton et al.'s (2008) argumentation theory, which recognizes that legal reasoning often involves defeasible inference, where multiple conclusions can be warranted given the same premises. Our system's lower confidence in interpretive entities mirrors the structure of legal argumentation itself, acknowledging that reasonable legal minds may differ in characterization while agreeing on objective facts.

5.1 Practical implications for Indonesian legal systems and lessons from other jurisdiction

Graph-based legal models introduce a new level of transparency, efficiency, and accountability to Indonesia's judicial system (Ashley 2019; Pan et al. 2024). Each prediction can be traced to specific nodes and relations in the graph, making reasoning steps explicit (Deeks 2019). For instance, a model predicting a heavy sentence can be explained by linking factual elements such as drug type, quantity exceeding 50 kg, corroborating evidence, and the application of Article 114(2), which prescribes a five-to-twenty-year range. This interpretability distinguishes graph-based systems from full-text models, which process thousands of words without revealing which features drive their predictions. Such transparency benefits judges, defense attorneys, and researchers by allowing them to verify reasoning, contest improper factors, and examine systemic bias or inconsistency.

Knowledge graphs also enable consistency analysis across time and geography (Castano et al. 2022; Leitner et al. 2019). They enable a systematic comparison of structurally similar cases that yield divergent sentences, helping to determine whether the differences reflect legitimate discretion or unequal treatment. For example, they can reveal whether narcotic sentences in Jakarta are consistently harsher than those in Yogyakarta or whether sentencing practices evolve following legislative reforms or Supreme Court rulings. By encoding legal relations rather than text, graphs support longitudinal studies that trace how judicial reasoning adapts to legal and social change, something impossible with unstructured documents whose format and phrasing shift over time.

The framework further enhances the efficiency and reusability. Once trained, a single extraction pipeline generates structured data that supports multiple applications, from sentence prediction and case retrieval to statutory compliance and consistency review. Although developing the system requires substantial initial investment, approximately 400 person-hours of annotation and engineering, it can process over a thousand cases per day once operational, creating a growing and reusable legal knowledge base. This structured foundation allows law students, practitioners, and researchers to query cases such as "narcotics trafficking between fifty and one hundred kilograms proven by surveillance in Jakarta, 2020–2023," retrieving comparable decisions instantly.

This model also holds value for education and professional development. Law students and junior judges can explore how variations in fact patterns affect sentencing, whereas prosecutors can reference historical trends when proposing penalties. Thus, technology enhances legal reasoning without displacing human judgment. However, implementation faces practical barriers. Many Indonesian district courts still lack adequate digital infrastructure and technical capacity (Bhakti 2016; Ginting, 2020; Napitupulu, 2022). A gradual rollout beginning with technologically ready jurisdictions such as Jakarta and Surabaya would allow the demonstration of tangible benefits such as greater transparency, faster access to precedents, and better resource management before nationwide adoption. Collaboration with law schools can help to cultivate professionals capable of interpreting and managing these systems responsibly.

Beyond Indonesia, the approach offers lessons for other Civil Law systems, such as Thailand, Vietnam, Malaysia, and the Philippines, where judicial texts share similar challenges of redundancy and inconsistent formatting (Ginting, 2020). The relevance-filtering method, which achieved nearly perfect accuracy and increased the extraction performance from 67 to 84%, is particularly transferable (Ahmad et al. 2020; Bhattacharya et al. 2019). This ensures that models learn from substantive content rather than procedural repetition, making structured representation feasible even in data-poor jurisdictions.

Evaluation should not stop with technical extraction metrics (Nuranti and Yulianti 2020). Traditional indicators, such as precision and F1-score, show mechanical accuracy, but not practical value. The ability of knowledge graphs to predict sentences with 85% accuracy, compared to 26 to 28% for full-text baselines, demonstrates that they capture legally meaningful patterns. Such extrinsic validation, testing whether extracted data truly support legal reasoning tasks, should become the benchmark for future legal NLP studies.

5.2 Limitations and bias concerns

The model predicts sentence length with 85% accuracy, but accuracy is not a measure of fairness. Machine learning reflects historical patterns rather than normative justice (Ahmad et al. 2020; Bhattacharya et al. 2019). If past decisions embed socioeconomic privilege, regional disparity, or institutional favoritism, the model may reproduce and even amplify these biases. Knowledge graphs do not eliminate bias, but they make it visible. By structuring cases around facts, evidence, and statutes, they enable the detection of disparities after controlling for legally relevant factors. Correcting inequality, however, requires institutional safeguards and formal fairness audits rather than technical fixes alone.

Algorithmic fairness auditing is therefore essential. Graph-based systems support comparative disparity analysis, counterfactual testing, subgroup performance evaluation, and judicial review of flagged cases. Human oversight ensures that fairness assessment remains a dialogue between empirical evidence and legal judgment. Institutional commitment is critical: courts must assign responsibility for monitoring bias, investigating disparities, and updating models. Without governance, AI risks reinforcing inequality under a veneer of objectivity. The framework should support judicial reasoning, not automate sentencing, since predictions reflect historical practice rather than statutory proportionality or human rights principles. Contestability is equally important. Defendants must be able to challenge extracted facts and model outputs (Deeks 2019; Marcos 2025). Each prediction should disclose its underlying graph triples, analogous cases, and confidence levels, providing auditability that opaque systems often fail to offer (Deroy et al. 2025).

The model has clear limitations. It mirrors existing judicial patterns, omits subjective factors such as intent or remorse, and covers only narcotics and corruption cases. Broader validation and integration with statutory knowledge graphs are needed. The study also does not benchmark modern LLMs. Future work should evaluate models across extraction accuracy, auditability, computational cost, consistency, hallucination rate, and robustness. Despite these constraints, the study shows that structured,

expert-informed representations can enhance transparency, consistency, and efficiency without displacing human judgment (Chen et al. 2024; Ershov 2023; Pan et al. 2024).

5.3 Future research directions

Future research can extend this framework in five connected directions that deepen its analytical power and strengthen its policy relevance. The first direction is the integration of statutory and decisional knowledge graphs, which would link what courts decide with what the law prescribes (Hoekstra et al. 2009; Santosh et al. 2024). Statutory graphs, where articles, offense elements, aggravating factors and penalties are modeled as linked nodes, would allow systematic comparison between legislative intent and judicial outcomes. Aligning the two graphs would show whether courts apply aggravating provisions consistently, whether non-statutory considerations shape sentencing, and how interpretations shift over time. This approach would provide an empirical basis for evaluating judicial discretion and supporting legal reform.

The second direction concerns multijurisdictional comparison, which can reveal whether sentencing varies across regions (Filtz et al. 2021; Kaltenboeck et al. 2022). Extending the framework to multiple provinces, for example by comparing Jakarta with Yogyakarta or resource-rich regions with administrative ones, follows comparative work in other legal systems (Leitner et al. 2019). Key challenges include harmonizing linguistic differences across courts, correcting for different case compositions and separating judicial behavior from contextual influences. Graph-based stratification and causal inference can help isolate genuine jurisdictional disparities and determine whether Indonesia's courts function as a coherent national system or a collection of diverse regional practices.

The third direction is temporal analysis. Treating court decisions as time-series data makes it possible to examine how judicial reasoning adapts to statutory reforms or Supreme Court rulings. Researchers can investigate whether sentencing changed after the 2009 narcotics law revision or how quickly lower courts adopted new precedents. Graph structures maintain stable units of comparison across time and therefore enable clearer analysis of how legal norms spread through the judiciary. The fourth direction is integration with argumentation theory (Alexy 2010; Prakken 2010; Walton et al. 2008). Embedding warrants, rebuttals and supporting authorities within graphs would model the reasoning pathways courts follow from facts to conclusions. These argumentation graphs could support judicial training, appellate review and consistency assessment, and can be aligned with statutory graphs to evaluate whether decisions rest on the appropriate legal provisions.

The fifth direction is the development of standards for fairness auditing in legal AI. Fairness should be assessed using both computational metrics and normative principles that are jointly defined by technologists, legal scholars and judicial actors. Finally, the framework should be expanded to include additional offense types such as fraud, embezzlement, assault and property crimes. This will test whether the hybrid extraction approach generalizes across domains with different evidentiary structures and legal requirements (Ahmad et al. 2020; Bhattacharya et al. 2019). These direc-

tions collectively outline a roadmap toward a more rigorous, transparent and accountable legal AI ecosystem.

6 Conclusion

This study shows that hybrid-extracted knowledge graphs can effectively represent Indonesian court decisions and improve downstream legal analytics. It contributes three things to legal informatics: a framework for processing noisy legal documents, empirical evidence that structured representations outperform full-text methods, and a replicable model for building legal AI in resource-constrained civil-law environments.

First, the study develops a hybrid extraction pipeline that pairs rule-based methods for structured sections with machine-learning models for narrative text. Applied to 9,109 Indonesian criminal decisions, it achieved an average F1-score of 91.9% for structured entities and 82–86% for unstructured ones. A major driver of performance was relevance filtering, which raised accuracy by 17.39 points (from 67% to 84.39%) by removing procedural boilerplate that dominates Indonesian court documents. Extracting sparse, legally salient content proved essential and offers a transferable solution for other civil-law jurisdictions with redundant or inconsistent formatting.

Second, the study empirically shows that knowledge graphs capture legal reasoning more reliably than full-text approaches. In the sentence-length prediction task, a proxy for proportionality analysis, graph-based models reached 85% test accuracy, whereas full-text baselines overfitted heavily, dropping from 95 to 97% training accuracy to 26–28% on test data. This 56.9-point gap indicates that judicial reasoning is fundamentally structured and compositional, arising from the interaction of statutes, facts, and evidentiary standards—patterns that symbolic representations model far better than unstructured text.

Third, the study offers a replicable framework for legal AI in low-resource settings. Strong results were achieved with only 200 expert-annotated documents, locally trained models, and cross-lingual transfer from English legal corpora such as LegalBERT. This demonstrates that capable legal AI does not require massive proprietary models or large annotation budgets and can be generalized to other civil-law systems with semi-structured documents and limited digital infrastructure.

Finally, the findings illuminate the epistemology of legal facts. Inter-annotator agreement is high for objective entities (≥ 0.90) but substantially lower for interpretive ones (≈ 0.52), reflecting genuine legal ambiguity rather than annotation error. Models should therefore flag low-confidence outputs rather than impose artificial certainty. Making contested zones visible aligns with argumentation theory's view of law as defeasible reasoning, where multiple conclusions can legitimately stem from the same facts.

Appendix A: Feature extraction from pdf documents

This appendix describes the supplementary pipeline developed to enable fully automated processing from raw PDF documents. While the main study utilized pre-extracted features from the Indo-Law dataset, this pipeline demonstrates a viable path toward end-to-end automation from original court decision PDFs.

A.1 Document collection and conversion

A total of 702 PDF files were re-downloaded from available sources and converted into text format using the pdfminer.high_level module. This tool was selected for its improved accuracy in representing the textual content of legal documents compared to alternative PDF extraction methods.

A.2 Text data preprocessing

Raw text extracted from PDFs underwent a series of preprocessing steps to ensure data quality and consistency:

- Removal of watermarks, headers, and footers.
- Elimination of superfluous punctuation.
- Deletion of records with missing values.

These preprocessing steps were essential for preparing clean text suitable for subsequent document segmentation and feature extraction.

A.3 Document segmentation methodology

The segmentation process aimed to predict paragraph-level segments from the original text and assign appropriate labels corresponding to four key features defined in the Indo-Law dataset: putusan.kepala_putusan, putusan.amar_putusan, putusan.fakta, and putusan.penutup.

A.3.1 Chunk size determination based on legal conventions

Indonesian court decisions follow a standardized structure in which different sections have predictable lengths that reflect their legal function. Chunk sizes were determined based on these legal document conventions rather than as arbitrary hyperparameters:

- Kepala_putusan (10 lines): Document headers contain case metadata including case number, defendant identity, and court information. According to SEMA 2022 specifications, this section must be concise, typically spanning 8–12 lines. The 10-line chunk size captures this standard header format.
- Amar_putusan (10 lines): Verdicts follow formulaic language specified by procedural law. The verdict statement begins with "MENGADILI" and enumerates decisions point by point. Standard verdicts contain 8–12 enumerated points, fit-

ting within 10 lines.

- Penutup (10 lines): Closing sections list judicial officers (judges, prosecutors, clerks) and their decision dates. This administrative information follows a fixed template, consistently spanning 8–12 lines.
- Fakta (40 lines): Fact sections exhibited the greatest structural variation. Simple cases may contain fact sections of approximately 500 words (roughly 25 lines), while complex cases may exceed 15,000 words (750+lines). The 40-line chunk represents a meaningful unit of factual narrative, corresponding to approximately 800-1,000 words. This size typically captures one complete factual episode in legal storytelling, where each episode describes a distinct criminal act, witness account, or evidentiary presentation following chronological or thematic organization.

Chunk sizes reflect functional units in legal discourse rather than arbitrary parameters. Templatic sections (headers, verdicts, and closings) have institutionally defined lengths. Facts sections require larger chunks because legal narratives are organized in episodes—a 40-line chunk provides sufficient context to understand a complete factual claim with supporting evidence and legal characterization. Smaller chunks fragment narratives and lose coherence, while larger chunks introduce noise from unrelated episodes.

Several chunk size combinations were tested: (10,10,30,10), (8,10,25,8), and (10,15,40,15) for (kepala_putusan, amar_putusan, fakta, penutup). Experimental results demonstrated that the combination (10,10,40,10) provided optimal accuracy. (see Figure 9).

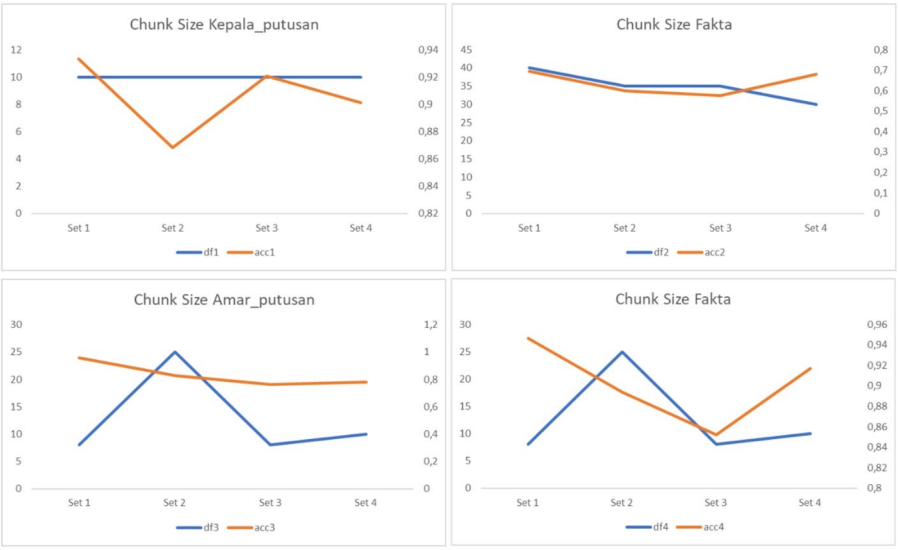


Fig. 9 Chunk size comparison

A.3.2 Chunk labeling procedure

For each document, the attributes putusan.@id and putusan.@sub_klasifikasi were retrieved directly from the Indo-Law dataset. Labels for remaining features were assigned to individual text chunks based on their semantic similarity to corresponding features in the Indo-Law dataset. For instance, the chunk exhibiting the highest similarity to the reference data for putusan.kepala_putusan was labelled accordingly. Chunks with the same label were subsequently aggregated to form coherent paragraphs.

The extracted data is used as training dataset to train BERT-base Model. The result from train and test is used as combined datasets in four different models related to putusan.kepala_putusan, putusan.amar_putusan, putusan.fakta, and putusan.penutup. This Models are expected to duplicate the same process for producing the Indo-Law dataset and prepare the upcoming new data from pdf files.

A.4 Model architecture and training

A fine-tuned IndoBERT model (indobenchmark/indobert-base-p1) was employed for document segmentation. This model architecture consists of 12 transformer layers with a 768-dimensional hidden state and has demonstrated strong performance in name and entity recognition tasks for corruption court documents.

Training Procedure: Text from the dataset was tokenized and wrapped into a custom PyTorch dataset. Training utilized DataLoader for batch processing with shuffled training data and employed the AdamW optimizer with a learning rate of 1×10^{-5}.

A.5 Results and performance evaluation

The optimal configuration combined a similarity threshold of 80% with chunk sizes of (10,10,40,10). Model performance varied across document sections:

- Kepala_putusan: 93.3% F1-score
- Amar_putusan: 88.1% F1-score
- Penutup: 91.7% F1-score
- Fakta: 69.1% F1-score

These figures correspond to rows d1 (kepala_putusan, 93.3%), d2 (fakta, 69.1%), d3 (amar_putusan, 88.1%), and d4 (penutup, 91.7%) in Table 12 below, ordered by document section rather than by performance. The lower performance on fakta sections reflects their inherent challenges: highest structural variation, boundary ambiguity between facts and legal reasoning, and extreme length variation across cases. Despite this limitation, the results demonstrate a viable path toward end-to-end automation from raw PDF documents.

For the main research presented in this study, analysis focused on data already extracted and curated in the Indo-Law dataset, ensuring consistency and reliability of features used for downstream tasks.

Table 12 Experiment setup for feature extraction

Setup		Acc	Prec	Rec	F1
d1	Train	93,32%	87,08%	93,32%	90,09%
	Test	93,39%	87,22%	93,39%	90,20%
d2	Train	69,58%	48,42%	69,58%	57,10%
	Test	69,16%	47,83%	69,16%	56,55%
d3	Train	95,95%	92,06%	95,95%	93,97%
	Test	96,52%	93,17%	96,52%	94,81%
d4	Train	94,67%	89,62%	94,67%	92,08%
	Test	93,37%	87,18%	93,37%	90,17%

Note: Rows d1–d4 correspond to document sections: d1 = kepala_putusan (document header), d2=fakta (case facts), d3 = amar_putusan (verdict), d4 = penutup (closing). Performance differences across sections reflect their structural complexity, with d2 (fakta) showing lower scores due to extreme length variation and boundary ambiguity

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Data availability The dataset used in this study is publicly available and can be accessed at https://doi.org/10.6084/m9.figshare.30510518.v1.

Declarations

Ethical statement This study did not involve human participants, personal data, or any form of experimentation on human subjects. All analyses were conducted using publicly available or secondary data sources, and no ethical approval was required in accordance with institutional and journal guidelines.

Competing interests The authors declare no competing interests.

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